International Association of Oil & Gas Producers

REPORT 2024sh | JUNE 2025

DATA SERIES

Safety performance indicators – 2024 data – High potential event reports

Acknowledgements

IOGP thanks those companies that have participated in the data collection programme.

This report was produced by the Safety Committee.

Feedback

IOGP welcomes feedback on our reports: publications@iogp.org

REPORT
2024sh

JUNE
2025

DATA SERIES

Safety performance indicators – 2024 data – High potential event reports

Revision history

VERSION	DATE	AMENDMENTS
1.0	June 2025	First release

Contents

AFRICA ONSHORE

DATE: 24 Oct 2024
COUNTRY: Gabon
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Transport – Land
PRIMARY LIFE-SAVING RULE: Driving
SECONARY LIFE-SAVING RULE: Bypassing safety controls
NARRATIVE:

As a convoy moved forward, the driver lost sight of the lorry, loaded with 25 tonnes of cement. It had been about 100 metres ahead of him. He decided to turn back. When he saw the lorry, it had overturned in a ravine. The driver was found a long time later, having been rescued by passers-by.

WHAT WENT WRONG:

Motor vehicle accident – a truck towing a cement tanker turned over.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The driver was under time pressure not to arrive at night, since night driving is not allowed on site. Revision of the convoy procedure.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Work or motion at improper speed
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

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DATE: 13 Jun 2024
COUNTRY: Gabon
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Transport – Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Bypassing safety controls
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While unloading the barge, a sailor employed by a contractor was injured by a tensioner connected to a steel chain attached to a trailer. The tensioner suddenly broke, causing injuries to the sailor's lips, the loss of a tooth, and unstable teeth.

WHAT WENT WRONG:

A sub-contractor (IP) struck in the face by a tensioned chain released when opening a lever turnbuckle. The incident resulted in an MTC involving 6 stitches, a lost tooth, and 2 unstable teeth.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The stowing and undocking procedure must always be followed every time.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 09 May 2024
COUNTRY: Ghana
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Unspecified – other
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During a heavy rainstorm, three aluminium composite panels/cladding (component of the external cladding material for the office building) dislodged and dropped. The three panels, each weighing 9.8kgs and measuring 1.11x1.49m, dropped from about 13 meters onto the canopy at the entrance to the reception of the building. There were no personnel in line of fire, hence no harm or injury incurred.

WHAT WENT WRONG:

Previous investigations revealed that no consideration was given to the potential for panels falling during design and installation. Furthermore, subsequent structural integrity assessments did not clearly establish the mode of failure and effective mitigation measures to prevent recurrence.

Additionally, in cases where potentially falling aluminium composite panels have been reported, remedial actions to remove and reinstall such panels are not promptly executed, despite the high potential impact of a dropped panel.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Based on the critical factors and root causes identified above, the following corrective actions have been proposed to address the underlying issues and prevent recurrence:

1. Thoroughly inspect and repair all weak/loosened panels to maintain the structural integrity and safety of the building (i.e., implement effective, timely corrective actions for the uncorrected dropped panel issue).
2. A qualified structural engineering company will be recommended to establish the mode of failure and recommend effective controls for consideration.
3. Track and implement all the recommendations from the Structural Integrity assessment report(s) on the Building.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

AFRICA ONSHORE

DATE: **15 Nov 2024**
COUNTRY: **Kenya**
FUNCTION: Production
CAUSE: Exposure electrical
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation

NARRATIVE:

A night guard carrying out routine security patrols noticed a sparking light near the critical containment area at an onshore camping base. The guards had been patrolling the area during their normal security patrols and during these patrols it has not been observed before. The guard reported the incident to the construction supervisor. The team went to the area and inspected area where the flashing light was seen coming from. It was discovered that the light was as a result of a live electrical cable that was sparking. The cable had been installed to provide power to the EX-rated lighting at the process area during the operations.

It appears that after partial decommissioning of the sites, the EX-rated system was left as built to save for power connections. There were no injuries or damages that were reported.

WHAT WENT WRONG:

It was observed that the 'as built' drawing includes the electrical cable for the lighting, which was not terminated following the decommissioning of site equipment. During the camp acceptance inspection in June 2022, it appears that the underground cable was not noted/flagged, and the camp was powered after the acceptance inspection was completed and signed off.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Physically Isolate(terminate) the cable from the distribution board.
- A survey to be conducted to detect underground cables using ground penetrating radar.

CAUSAL FACTORS:
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: **26 Nov 2024**
COUNTRY: **Libya**
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

The water truck driver attempted to perform a maintenance activity under the truck, while the engine was running and the power take off (PTO) was engaged. During the maintenance activity, the driver's arm was caught between parts of the engine, resulting in a severe injury. The injury was serious enough that the arm was amputated as a consequence of being trapped.

WHAT WENT WRONG:

- Lack of energy isolation.

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- Lack of risk assessment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Follow the safety basic rules regarding energy isolation and risk assessment.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

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DATE: 29 Jun 2024
COUNTRY: Angola
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

An employee burned his right leg, in the region of the knee, due to hot water projection. At around 11:45 am, during the operation to replace the motorized valve in the hydro cyclone unit, hot water at a temperature of 95ºC and a pressure of 5bar, coming from the same line, hit the employee in the region of his right knee, causing him a first-degree burn.

WHAT WENT WRONG:

Communication failure before and during operation, non-compliance with the hydro cyclone unit isolation process procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Before starting the work, the person responsible for the area must confirm the availability of the equipment; continuous training of the equipment isolation system (logout/tagout); awareness in the use of electrical process insulation certificates.

CAUSAL FACTORS:
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement

DATE: 11 Apr 2024
COUNTRY: Ghana
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During crane operation, the port crane auxiliary winch was being used to lift a waste skip containing concrete waste materials weighing 6.8 tonnes from the main deck onto the laydown area. The crane operator lifted the load approximately 7 metres above the main deck and in the process of slewing towards the port laydown area, the load on the auxiliary hook suddenly descended in a rapid uncontrolled manner. The load, during its descent, struck a light fitting prior to landing on the port main deck in an upright and stable position. No unreeving of auxiliary wire rope was observed, and it continued to remain under tension.

WHAT WENT WRONG:

Independent from the non-technical investigation, a technical report has been received and in summary the investigation confirmed that the crane auxiliary motor counter balance valve (CBV) was leaking and was damaged,

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and the motor was found to have high case drain leakage of hydraulic oil, possibly due to scoring that was observed on the hydraulic motor port plate. It is likely these two factors combined resulted in the load dropping.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

With a new motor, new CBV fitted, new brake installed, the fault could not be recreated. New hydraulic oil was cleaned and installed into the crane. A full set of assurance checks carried out including a proof load test.

The following are longer term recommendations.

- Install pressure compensation sandwich valves under each direction control valve, to prevent the increase in flow and in turn increase in pressure in the auxiliary and slew circuits being shared between the operating functions of the crane, when operating multiple functions at the same time.
- Install a flow control valve in the load sense line (as per the Starboard crane setup). This will dampen the response from the pump, e.g., when aux lowering and bringing in the boom raise.
- Complete a review of the CBV sizing. It may be beneficial to reduce the flow capacity, although the next size down is a 240l/min which may be too restrictive, therefore it may also require a new manifold to accommodate dual CBVs.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 04 Feb 2024
COUNTRY: Ghana
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

The port crane was in operation for material backload onto a supply boat. At about 15:58hrs, the crane operator lifted a load (half height container with 5 IBCs containing neutralized acid/diesel mixture) and in the process of slewing towards overside of the FPSO, the auxiliary hoist wire rope parted; dropping the load from an estimated height of 6.6m. The load made contact with two containers on the main deck close to the port laydown and fell overboard into the sea. Four (4) of IBCs containing neutralized acid and the half height container sank to the seabed leaving 1 IBC afloat the sea surface. The operation was immediately halted. There was no personal injury, but traces of sheen observed on recovery of the floating IBC.

WHAT WENT WRONG:

Root Causes includes:

- Maintenance – Significant wear on the auxiliary rope not picked up during routine inspection.
- Competency gap – Hydraulic supervisors/crane inspectors had not undergone specific training on crane wire rope inspections.
- Inadequate development of procedures (maintenance strategy).
- Lack of assessment of risk exposure.
- Boom Rest Protection was inadequate.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Some of the actions recommended are:

- Update vendor training requirement in the contracts, which should include wire rope training for persons involved in crane inspections on the FPSO.
- Review and update crane maintenance strategy to include auxiliary rope boom rest protection, frequency of rope change-out, suitable rope and sheath gauge to be used as a tool for measuring rope wear, etc.
- Install adequate auxiliary rope boom rest protection.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 31 Dec 2024
COUNTRY: Ghana
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A light fixture affixed to short length of cable tray with a combined weight of 14.45kg detached from its temporary securing mechanism and fell approximately 3.9 meters onto scaffolding below. There was one person in the line of fire. The light fixture made slight contact with the person's helmet and face shield before landing on the scaffolding platform.

However, there were no injuries to personnel. No visible damage observed on hard hat and face shield.

WHAT WENT WRONG:

The investigation team determined that "incorrect use of plastic cable ties for the intended purpose" and the "improperly routed light cable (temporary setup)" were critical factors contributing to the incident. Additionally, the team identified several root causes, including poor judgement, the absence of established policies and procedures, inadequate work planning, and insufficient skills/knowledge of reliable securing methods.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

To prevent re-occurrence, the investigation team recommends the following corrective actions:

- Prepare a one-page summary of the event and circulate for use at team briefs, town halls, and safety meetings to inform and educate the entire crew on the causal factors and reinforce expectations on dynamic risk assessment and dropped object hazard.
- Devise a training and awareness plan to educate all crew members on Reliable Securing, promoting thorough understanding and ensuring adherence to best practices based on the DROPS (Dropped Object Prevention Scheme) guidance document.
- Consider mobilising an external resource to facilitate training sessions for all crew to enhance awareness of DROPS principles and evaluate the effectiveness of existing dropped object controls and inspections, aligning DROPS best practice to our needs.

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CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: 27 Jul 2024
COUNTRY: Ghana
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During routine personnel rounds, a metal component of the Gas Turbine Generator (GTG) exhaust diffuser was found on the mezzanine deck near the exhaust stack of the GTGs. The exact conditions required (failure of strut assembly, size of section, Waste Heat Recovery Unit loading, overall turbine load) to eject the component could not be accurately known. A few weeks after the subject dropped object incident, GTG A turbine was removed for unplanned engine exchange due to a bearing failure. At this time, an opportunity to inspect the exhaust presented itself.

WHAT WENT WRONG:

The main root cause identified for the incident was the inadequate assessment of the potential failure due to the lack of designed facilities (access hatch) for inspection and maintenance of the GTG exhaust assembly. Other associated findings included the lack of a dedicated inspection plan for the exhaust area as there was no direction/clarity in the OEM procedure during opportunities when engine is removed and a suspected poor design/quality of materials from the original packager.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The following actions are proposed to address the findings:

- Develop a long-term repair strategy; either to upgrade or repair the existing exhaust diffusers. Include provisions as required for inspection.
- Adjust maintenance routines to clearly include inspection of exhaust whenever there is an engine removed.
- Obtain clarity on expectations from packager/manufacturer on extent of inspection required for exhaust components as well as comment on if there are deficiencies in the engineering and/or material selection from original packager.
- Make provisions to inspect GTG's exhaust assembly to determine the condition of internal components and be ready to react to findings (i.e., if any replacements are required).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 16 Apr 2024
COUNTRY: Ghana
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

An anomaly was observed during routine plant inspections. A large flat metal sheet was found resting on the forward refuge roof. Subsequent investigation revealed that this section was a cable tray cover dislodged from the pipe rack above the forward machinery, with a drop trajectory indicating its passage across a walkway. The cover weighed approximately 20kg, and its landing point on the forward refuge area roof was 13.356m from the pipe rack (original location).

WHAT WENT WRONG:

The root causes included inadequate inspection/monitoring practices coupled with insufficient risk exposure identification at design stage.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

To address these findings, the following actions are proposed:

- Review and enhance the current securing method for cable tray cover panels, incorporating a secondary securing mechanism. A sample of secondary securing mechanism shown on page 3. Sample picture picked.
- Develop a comprehensive inspection and maintenance strategy for cable tray cover panels – alternatively consider including inspection of cable trays in the general structural inspections or into the scope of the flare tip drone inspection.
- Develop a guidance document for checks required for areas at height. Guidance checks will be incorporated into the inspection process.

CAUSAL FACTORS:
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 08 Mar 2024
COUNTRY: Ghana
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

After access was granted for valve line up in preparation for methanol transfer adjacent to the flare stack, a piece of metal was found on the deck. The metal piece was picked up and incident reported to the production supervisor.

WHAT WENT WRONG:

The modified HP flare tip was designed like-for-like with an enhanced design and constructed as such to replace

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the old one which had a dislodged tulip in 2017. Original equipment manufacturer (OEM) did not advise on the potential for chattering as well as its impact on the HP flare tip's structural integrity even though it was clear that this had been experienced on similar models. If this had been communicated at the onset, then the no flare chattering ranges would have been instituted to mitigate plate failure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Review available flare tip remediation options and decide on the replacement plan.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

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DATE: 20 Jan 2024
COUNTRY: Nigeria
FUNCTION: Production
CAUSE: Aviation accident
ACTIVITY: Transport – Air
PRIMARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

On January 20, 2024, a commercial air transport flight encountered a flight upset, leading to a loss of control incident while enroute to an FPSO (offshore facility). The AW139 helicopter experienced extreme unusual flight attitudes 35 minutes into the flight, resulting in the detachment of three emergency push-out windows, minor injuries to three passengers, and minor damage to the aircraft. After experiencing extreme aircraft attitude changes and very high rates of descent down to only 306 ft above the sea surface, the crew regained control of the helicopter after 6:35 minutes and safely returned.

WHAT WENT WRONG:

Activation of the aircraft's electrical system gang bar to resolve an autopilot (AP) system failure, whilst in instrument meteorological conditions (IMC), with causal factors related to persistent aircraft system unreliability, a breakdown in crew cockpit procedures and insufficient aircraft system knowledge. These were further influenced by an organisational and leadership culture characterized by mistrust, resource limitations, and breakdowns in operational processes.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The resilience and performance of the approved aircraft type significantly contributed to the safe outcome.

The aircraft assessment process does not address optional service bulletin policy or assess the safety impact and risk associated with each service bulletin.

There is little understanding/awareness in the industry or by the OEM of the inherent weakness and possibility for normalization of incorrect use of the aircraft's gang bar design, with numerous operators using it outside its design intent. With the AW139 making up over 40% of organisation's helicopter exposure, this needs to be addressed globally.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

ASIA/AUSTRALASIA ONSHORE

DATE: 04 Oct 2024
COUNTRY: Australia
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During insulation activities on Heat Exchanger. A plastic bag with rock wool dropped approx. 8 meters with a load weight of approx. 16kg.

The work group had secured the bag with sisal rope at the top. During lifting the load, the knot appears to have failed. (no certified lifting bag used).

WHAT WENT WRONG:

- Deviation from process to lift manually with a rope, without a properly rigged/rated lifting bag or exclusion zone in place.
- The work area had changed and the way the lift might usually have been undertaken (chaining/crane) was not possible at this time.
- Self-imposed pressure to get the task completed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Training Team Members – Gin Wheel (pulley).
- Including Reminder Notices, Induction coverage, add to Risk Assessment.
- Acquiring and supplying additional lifting bags and the creation of Gin Wheel Kits for end user ease.
- Talking about it. Conduct information sessions to be run for team to aid understanding and awareness of lifting requirements and expectations.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA ONSHORE

DATE: 15 Jun 2024
COUNTRY: China
FUNCTION: Construction
CAUSE: Pressure release
ACTIVITY: Excavation, trenching, ground disturbance
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

Gas pipe being damaged by excavator while construction project was undergoing. Luckily, workers observed immediately and fixed it soon. No worse result happened.

WHAT WENT WRONG:

JSA was lack of deep thinking.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

JSA shall be done well before the job.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: 17 Jul 2024
COUNTRY: India
FUNCTION: Unspecified
CAUSE: Falls from height
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Working at height

NARRATIVE:

Falling of the Topman while coming down through TEED

WHAT WENT WRONG:

Turn buckle sheared from the threaded portion

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Usage of TEED during emergency; Material checking of the buckle

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA ONSHORE

DATE: 25 Apr 2024
COUNTRY: Indonesia
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Flash Fire in Maintenance of Ignition Flare Line at SPU-A Jatibarang Field

WHAT WENT WRONG:

- There was no positive isolation for fuel line when maintenance took place.
- There was no procedure to change the panel from automatic to manual when maintenance took place.
- The supervisor did not come to the location.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- The needed of isolation must be decided before the job.
- Identification of hazard in risk assessment must be completed.
- Need procedure to guide specific job.
- Supervision.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 30 Oct 2024
COUNTRY: Japan
FUNCTION: Unspecified
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Three employees, two for rigging and one crane operator, were lifting a bundle of 58 sucker rods from the bed of a 26-ton trailer with a 5-ton overhead crane and the IP (external driver) had his hands on the load (each bundle 7.8m long, 0.75m wide, and 0.29m high, with the load floor height at approximately 2.6 meters and a total weight of about 1.1 tons). After lifting the load to approximately 0.7 meters, they began to slowly move it towards the back of the vehicle.

The hook's safety latch broke, causing one of the two slings to detach from the crane's hook, which resulted in the load falling. At that moment, the IP had his hands on the load and was located underneath it, resulting in them getting caught and injured between the falling load and the remaining cargo.

Additionally, the falling load got caught on a deck board and did not fall completely.

ASIA/AUSTRALASIA ONSHORE

WHAT WENT WRONG:

- Warehouse information and characteristics were not listed.
- No additional training (retraining) was conducted after obtaining rigging operation qualification acquisition; there were no supervisors present, nor guidance provided from others.
- There was a lack of personnel who could assess the risks and decide whether unloading was possible, including necessary schedule changes.

Although there were documented tool box meeting TBM procedures, they were not followed.

- The safety work standard document states that "long items like tubing and casing should use tag line ropes," but it was not recognized that the tag line ropes were needed for other items.
- Due to the extended trailer size, not all cargo could fit inside the warehouse; after the ground release, the cargo had to be moved westward inside the warehouse.
- TBM did not provide proper staffing instructions or guidance.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implement education on rigging practices using materials such as DVDs and texts.
- Enrol in retraining courses related to rigging.
- Create and regularly update a comprehensive list of each warehouse's information and characteristics to assist in deciding equipment delivery locations.
- Develop and continuously practice using check cards during TBM, updating them as necessary.
- During TBM, use check cards to confirm proper division of roles and personnel allocation, and wear chin straps on helmets specific to each role while performing tasks.
- The supervisor should ensure the safety of personnel from a position that allows them to see the entire area before and after lifting object.
- Update the safety work standard document to mandate the use of tag line ropes for all lifting operations.
- Conduct readings of safety work standard documents and lifting manuals during morning meetings and pre-work meetings.
- Implement LSR education. Clearly define and educate policies for resuming operations after an incident.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

ASIA/AUSTRALASIA ONSHORE

DATE: 26 Aug 2024
COUNTRY: New Zealand
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

IP had advised his colleague he was going to carry out continuity testing on the electric submersible pump (ESP) cable, which was on the hydraulically energised spooler. The spooler was not in use at this time.

When carrying out the continuity testing, the conductor reading was low and the cable was cleaned with electrical cleaner. The next step was to test the earth cable. This required reaching into the cable drum between the beams of the spooler frame.

As the IP reached into the cable drum to grab the earth cable end and carry out the test, an operator, who was standing at the front of the spooler (unaware that continuity testing was being carried out and could not see the IP from his vantage point) operated the spooler to raise the ESP cable.

The IP yelled out and the spooler rotation was stopped however, the IP's right hand had been caught between the rotating drum and the spooler frame.

WHAT WENT WRONG:

- IP did not follow the procedures for this task and therefore the spooler was not isolated.
- Specific hazard associated with the ESP spooler cable continuity tests was not identified in the issued permit to work.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review and update work over unit permits.

Communication of hazards in toolbox meetings.

- Understanding of isolation process from all operators and involved parties.
- Review electrical check procedures.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

DATE: 24 Mar 2024
COUNTRY: Papua New Guinea
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Safe mechanical lifting

NARRATIVE:

Derrickman impacted by rotary table bushing inserts after sudden energy release during removal procedure.

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WHAT WENT WRONG:

Driller's side tugger line eye thimble became caught on the top drive frame. With increased tension applied to the winch line, the thimble then dislodged from the top drive, transferring the stored energy to the bushing insert, resulting in the uncontrolled movement.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Eliminate potential snag points from tugger rigging.
- Replan task to remove personnel from red zone.

Define trigger point for stuck bushings in job safety analysis (JSA).

- Install weight indicator on tugger to control tension.
- Investigate alternate bushing pulling equipment.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 08 Aug 2024
COUNTRY: Philippines
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Office, warehouse, accommodation, catering
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A portion of hardened concrete cement plaster from a crack portion of the wall from living quarters suddenly fell. The cement plaster has an estimated weight of 14 kg, and a dimension of 3x100x25 cm which after falling from an estimated height of 6.5 meters, was broken into pieces. It was also noticed that before the piece of concrete cement fell on the ground, it hit a stainless tube on the 2nd floor resulting in its deformation. Based on the dropped object consequence calculator, an incident will result in death resulting from an injury or trauma.

The area was immediately barricaded and all vehicles near the area were relocated. Sighting of concrete cement wall crack that might also fall was also barricaded and personnel were re-routed to access the living quarters.

WHAT WENT WRONG:

- Aging asset.
- Thick cement plaster.
- Inspection and maintenance is reactive in nature.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Improve on maintenance of aging assets.
- Strengthen programs for people to be vigilant for potential falling objects like structural fixtures (e.g., cement plasters) and coconuts.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

ASIA/AUSTRALASIA ONSHORE

DATE: 17 Feb 2024
COUNTRY: Thailand
FUNCTION: Construction
CAUSE: Exposure electrical
ACTIVITY: Excavation, trenching, ground disturbance
PRIMARY LIFE-SAVING RULE: Work authorization
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

High voltage underground cable was damaged from backhoe during excavation.

WHAT WENT WRONG:

Unclear excavation checklist – the requirement of attachment document for excavation work template was put in "Checkbox" not set as mandatory box. Contractor did not have process to verify the competency of their Task Supervisor (TS) and allow new TS who was not verified for their competency to work alone and contractor assigned Task Supervisor who did not have license to perform excavation work.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Revise document checklist to add more detail of required document to prevent confusing of the document that requires to attached for the excavation Permit to Work.
- Develop the work instructions and criteria for Task Supervisor competency evaluation.
- Register the list of PTTEP approved Task Supervisor for excavation work.
- Company to enforce Contractor for providing competence workforce to Company.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 03 Jan 2024
COUNTRY: Thailand
FUNCTION: Unspecified
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

On January 3, 2024 at approximately 03:30 AM, a lifting operation was started to transfer an open-top container (Max gross weight 10 tons), which was used for lifting sling storage, from the top of the firing line deck to the main deck using a crawler crane.

The container was lifted about 1 foot above the deck using the auxiliary block of the crawler crane. The crane operator noticed that the load indicator showed that the weight of the container had reached the specified safe working load (SWL) limit of 10 tons, but the crane alarm set point of the auxiliary block, which was set at 13.5 tons, was not activated. The operator stopped the lifting operation immediately and informed the banksman to change to the main block, which had a SWL of 35 tons. While the crane operator was attempting to lower the container, the

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auxiliary crane wire broke, causing the auxiliary block of the crawler crane to fall into the open-top container, and the container suddenly landed back onto the position where it was located before it was lifted.

There were no personnel injured from this event.

Note: According to the initial verification it was found the actual weight of the open-top container is 15 tons. The drop distance of the auxiliary block to the open-top container is approximately 35 ft.

WHAT WENT WRONG:

N/A

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

N/A

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

ASIA/AUSTRALASIA OFFSHORE

DATE: 07 Oct 2024
COUNTRY: Australia
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During Vertical Lay System (VLS) winch maintenance, the technician stepped backward into an open ladder hatch resulting in a grazed right shin. The VLS was not being used for the scope of work at the time of the event. The crew member stepped/tripped into an open hatchway that was located on the raised platform that leads to the fixed ladder. Potential fall height to the lower level was up to 6 metres.

WHAT WENT WRONG:

- A VLS Technician left the ladder access/egress hatch in the open position. This is common practice to leave the hatch open for access/egress purpose due to the design (i.e., weight).
- IP assumed the hatch was closed and stepped back into the open hatch.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Poorly designed access ladder hatch covers created error traps (a condition that makes it more difficult to work safely and which increases the likelihood of errors – e.g., hatch not being closed).
- No assessment completed to identify access and egress methods or risks created via access method.
- Procedures and process not followed.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 01 Jan 2024
COUNTRY: Australia
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Transport – Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Pipelaying vessel was conducting pipelay operations on Dynamic Positioning (DP), when the DP system failed resulting in a loss of vessel position. The vessel moved astern under the action of the tensioners, and the DP operators (DPO) recovered control of the vessel position from the back-up DP console. During this movement, a section of pipeline was pushed through the bow hatch of the vessel and dropped to sea. Personnel in the working area were mustered and accounted for; no injuries occurred.

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WHAT WENT WRONG:

1. Equipment malfunction – malfunction of the DP utility panel resulted in the activation of the "Back-up Levers" command.

- Equipment parts defective

2. Human performance – human activation of the "back-up levers" command.

- Controls were not designed to prevent unintentional activation
- Controls did not provide feedback to confirm that panel was powered
- Poor communication meant that all relevant personnel were not aware that the panel was powered.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Perform robust Management Of Change when upgrading functionality of any Safety Critical Equipment to fully understand implications of the new system and the residual risks these may have to operations (i.e., assess need for single manual override functionality for all thrusters).

Conduct drills in key work areas considering different emergency scenarios, cause for evacuation, and appropriate response to build familiarisation with primary, secondary and tertiary evacuation routes.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 21 Jun 2024
COUNTRY: India
FUNCTION: Unspecified
CAUSE: Unspecified – Other
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Fire in battery room at unmanned platform.

WHAT WENT WRONG:

Non dissipation of the heat from batteries leading to fire.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Issue with the material.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate supervision

ASIA/AUSTRALASIA OFFSHORE

DATE: 17 Jun 2024
COUNTRY: Indonesia
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

While conducting routine patrol, production operator noticed hissing sound around production cooler on main deck. He then reported to central control room (CCR) operator via radio. Production operator and instrument technician proceed to main deck and confirmed gas leak was coming from 8" elbow outlet production cooler bay. Offshore installation manager (OIM) and production supervisor was informed by CCR operator. OIM instructed CCR operator to close all wells immediately.

No fix gas detector on main deck was activated. No gas cloud being observed. Wind speed was 10–13 knots, wind direction 200 degrees. All wells were shut in at 21:10hrs.

When the situation is safe, production technician found small hole at the 8" elbow of outlet production cooler with working pressure of 10 bar (147 psi). The leak hole diameter is 4.4 mm. Based on LOPC calculator, the estimate hydrocarbon release within 25 minutes duration is 45 kgs.

WHAT WENT WRONG:

Immediate cause: Plant/equipment malfunction/defect: pipe corrosion/erosion.

Root causes:

- PPIG incomplete or lacks sufficient guidance.
- Failure to adhere to PPIG/taking shortcuts.
- Weak individual or team accountability of PPIG.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Undertake a detailed review on sand management to optimize and implement improvements.

CAUSAL FACTORS:

No causal factors allocated.

DATE: 17 Feb 2024
COUNTRY: Malaysia
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Transport – Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During a hawser connection between the floating storage and offloading unit (FSO) and an offtake tanker, the messenger rope from the hawser became stuck on an assisting vessel's fender. The crew attempted to manoeuvre the assisting vessel away to release the rope but it snapped back with tension and struck the back of the IP's head. The IP fell forward and hit the vessel's structure (bulwark) which resulted in a facial laceration, jaw fracture, and dental trauma to several teeth.

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WHAT WENT WRONG:

Due to current and swell directions, the messenger rope got entangled with the vessel's tire fenders. As the vessel was clearing the area, the crew was instructed to clear the deck. A crew member (IP), looking for shelter, went under the crash rail, misjudging it as a safe place. The IP noticed tension in the hawser assembly and looked over the bulwark to investigate and at that moment the messenger rope lashed down on the back of the IP's helmet which caused him to fall forward and hit his face against the vessel's bulwark.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure job safety analysis (JSA) for this activity is revised to incorporate safe haven risk of entanglement.
- Reinforce toolbox talk requirements; new shift joiner must perform a toolbox talk prior to entry.
- Refresh all supervisors regarding their supervisory role awareness. Introduce and perform awareness on the requirement of a spotter.
- All parties involved must attend pre-offtake meeting and agree on execution plan (e.g., offtake standard work instructions).
- Tanker escape route, its identification, and activation protocol to be included in the offtake meeting's terms of reference.
- Revisit the FSO offtake Risk Assessment and JSA .
- Review the tire fender design for North Malay Basin operations and explore operational technology advancements for safer messenger rope connections to minimize entanglement risks.
- Incorporate safer hawser connection procedures and 'no slack' requirements into the Standard Work Instructions (SWI).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

DATE: 02 Sep 2024
COUNTRY: Philippines
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A beacon (dynamic positioning transponder) was being lifted in its lifting frame by the vessel crane in preparation for deploying overboard. As the load was being lifted, it began to swing out of control and the frames bridle master link disconnected from the positive locking remote operating vehicle (ROV) hook resulting in the lifting frame and maxi beacon (weighing 163kg) dropping 6m to the ships deck.

As per the DROPS calculator, this could have resulted in a fatality.

WHAT WENT WRONG:

- Inadequate communication of standard for lessons learned and incidents.
- Inadequate judgement and risk perception, in part due to:
  - Inadequate vessel specific experience in high sea states.
  - Poor risk perception in proximity to load being lifted.
  - Inadequate supervisory participation in pre-task briefings.

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- Ambiguous lift plan, due to different sections with conflicting descriptions.
- Inadequate equipment knowledge: any equipment that is not industry norm should be considered non-standard and should not expect crew and rigging inspectors to have in-depth knowledge of the equipment. Non-standard equipment should be specified in the scope along with original equipment manufacturer (OEM) instructions.
- Inadequacy in terms of having a mature culture of intervention and supervision within the site.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Lessons learned from incidents should be communicated properly. A register of these should be available for easy reference.
- Any equipment that is not industry norm should be considered non-standard and should not expect crew and rigging inspectors to have in-depth knowledge of the equipment. Non-standard equipment should be specified in the scope along with OEM instructions.
- Update lift plans to include notes for lifting light loads.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 06 Sep 2024
COUNTRY: Philippines
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During a maintenance activity on the fuel gas system, it was observed on a pressure transmitter that the pressure went down to around 16 bars then went up again to around 31.44 bars which is above the rating of the downstream pressure safety valve (PSV). The PSV was observed to have cycled 12 times over 7 minutes before the pressure on the fuel gas line declined to normal pressure at 22.5 bars. The fuel gas piping and header downstream was subjected to pressure of around 33 bars, higher than documented design rating.

WHAT WENT WRONG:

- PSV recalibration/recertification testing done at least once whereas API 576 RP recommends at least three pops.
- Leak on fittings due to under/over-tightened fittings.
- Leak on past seals due to wear and tear/improper installation.
- Excessive PSV null zone due to incorrect setting of adjuster bottom (position of supply seal relative to exhaust seal).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Follow PSV recalibration/recertification industry standards.
- There is a need to improve the PSV certification process and the competency requirements of the QA/QC personnel and how the QA/QC personnel understand the reference document in the test plan given.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

ASIA/AUSTRALASIA OFFSHORE

DATE: 22 Jul 2024
COUNTRY: Philippines
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

The general platform alarm (GPA) was activated due to a loud explosion sound heard. Upon checking by emergency response team (ERT), it was determined that a valve's body bolts ruptured causing the valve body to split into two.

The valve could cause dropped object injuries if its projectile's line of fire hit a person.

WHAT WENT WRONG:

- Chloride Stress Corrosion Cracking (Chloride SCC).
- Undefined Body Bolt Specification for Valves.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Body bolts should be specified in valve procurement to take into account the environment in which it will be subjected.
- Assessment of materials used should be done to prevent conditions that would lead to chloride stress corrosion cracking.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 09 Feb 2024
COUNTRY: Thailand
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

The crane exhaust muffler fell from the top deck to the stern of the supply boat during a lifting operation. No personal injuries or property damage occurred.

WHAT WENT WRONG:

Potential dropped object was identified but the interim corrective actions were not taken properly.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Prior to conduct lifting operation and drop survey, conduct pre-post lifting checklist, and crane component drop checklist.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

ASIA/AUSTRALASIA OFFSHORE

DATE: 07 May 2024
COUNTRY: Thailand
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Rich glycol liquid overflowed from reboiler still column then soaked inside the reboiler shell insulator and fire was ignited at glycol reboiler shell insulator.

WHAT WENT WRONG:

- No LALL and PAH on glycol surge tank and no LAH at glycol reboiler.
- Glycol reboiler insulation jacket has hole and incomplete seal.
- No routine check valve/vent line blockage verification.
- No specific procedure to alarm on high level at glycol reboiler.
- Lack of competence assurance and supervision.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Replace carbon filter/wet filter at ERPC with appropriate duration.
- Set up preventative maintenance program to verify glycol reboiler 1" vent line and check valve.
- Install LAH on glycol reboiler and PAH/LALL on glycol surge tank.
- Provide training for glycol skid operation and update operating procedure.
- Conduct process safety upset drill on glycol overspill case.
- Develop job safety analysis for glycol refilling task to identify risk mitigation before starting job.
- Revisit glycol reboiler insulation inspection and replacement.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement

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DATE: 22 Sep 2024
COUNTRY: Thailand
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Production operator observed light smoke without flame coming out from gas compressor vent fan stack.

WHAT WENT WRONG:

The lube oil was leaking from the lube oil tube and vaporized when touching the hot surface inside the enclosure.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Root causes are lack of proper leak test procedure and the re-used tube. Therefore, the investigation team recommends to change the lube oil supply tube of all applicable turbine model after its disassembly and develop the leak test procedure.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 07 Jun 2024
COUNTRY: Thailand
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

During an electrical & instrumentation job at top deck, gas suddenly leaked from a grease nipple of a 6" VB recycle of suction gas compressor. They then informed Supervisor/central control room to decrease speed of gas compressor and control shutdown. Then isolate and bled pressure of valves. At 09:45 hrs. completed bleeding pressure and no continue gas leak from grease nipple.

WHAT WENT WRONG:

Grease nipple of m6"VB recycle of suction gas compressor was cracked.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implement relief valve and grease nipples survey for all CPP, Phase-1, WHP manual valves. Replacement campaign in December 2024.
- Check condition of internal check valve and replace per condition in relief valve and grease nipples replacement campaign in December 2024. Use sealant injection new model with cap.
- Launch CPP manual valve greasing program (planned Q1 2025). Establish PM plan by revisiting manual valve criticality and implement risk-based maintenance.
- Phase I Unmanned (FEED Study). To expand F&G coverage for unmanned operation.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

EUROPE ONSHORE

DATE: 03 Apr 2024
COUNTRY: Austria
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

Blow out. During mounting, the borehole started to overflow.

The crew tried to secure the well – without success.

Alerting and mobilization of the necessary emergency personnel.

After installing two preventers, the well was secured and pumped dead with 27m³ of mud.

The well was regularly checked up overnight, and clean-up work was also carried out.

A Workover Rig was replaced, the well was additionally secured with a long stroke (LS) packer and the blowout preventer (BOP) system, including the Christmas tree lower part, was completely replaced.

WHAT WENT WRONG:

Insufficient fluid in the well bore.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Exact volume calculation of well bore sufficient fluid on location.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement

DATE: 18 Sep 2024
COUNTRY: Germany
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Two employees were busy assembling the bell nipple (spout) on the Blowout Preventer (BOP). In order to reinstall it after completion of wireline work in which a lock element was used, the bell nipple had to be pushed back onto the BOP with greater effort. When moving this component between 2 double T-beams in the substructure, the bell nipple detached from the supposed guide that was not intended for this purpose and fell onto the drill cellar cover. Both employees reflexively moved out of the danger area. In doing so, they were able to move backwards and avoid the main body of the component. The bell nipple hit a cover plate of the borehole cellar and turned with the spout in the direction of the injured employee. The injured employee was injured in the shin of his right leg.

After immediate first aid, the injured employee was transported by ambulance to the hospital.

A DROPS analysis using the DROPS calculator will be performed as soon as the specific data on the bell nipple is available.

EUROPE ONSHORE

The incident occurred shortly before the end of the shift (scheduled end of shift 18:00, time of the incident 17:50).

The employee concerned had started his shift at 12:00p.m due to the journey to the scheduled start of the shift.

WHAT WENT WRONG:

IMMEDIATE CAUSE:

Substandard conditions and information.

BASIC CAUSE:

Inadequate product/service design.

Inadequate communication and information.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Not available for reporting.

CAUSAL FACTORS:

No causal factors allocated.

DATE: 05 Feb 2024
COUNTRY: Germany
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During installation work on a sucker rod pump with a cherry picker, the working platform with an employee inside fell approximately 20ft to the ground.

WHAT WENT WRONG:

Pump jack started unexpectedly and struck the basket of the cherry picker – IP started cleaning, stopped to allow horsehead repositioning, and then resumed cleaning without verifying energy isolation.

- Repositioning horsehead requires removing energy isolation.
- Company requirement for removing energy isolation states "The area operator must confirm that no other planned or ongoing work requires any of the existing isolation points to remain isolated".
- Verifying isolation prior to working adjacent to equipment is a fundamental step in work management and should be a part of start work checks. There was a lack of recognition by the work team of the risk of continuing to work next to the pumpjack with no lock in place.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review scope and application of personal isolation process for work that involves multiple workers/tasks.
- Consider updating the global company management system – personal isolation section with the lessons learned from this incident.
- Share this incident with the global company management system network with guidance on proper use of personal isolation process.

EUROPE ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

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DATE: 18 Mar 2024
COUNTRY: Hungary
FUNCTION: Production
CAUSE: Exposure electrical
ACTIVITY: Excavation, trenching, ground disturbance
PRIMARY LIFE-SAVING RULE: Work authorization
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Contractor installed an earthing probe without consultation during the installation of experimental water purification equipment. During the installation, a live electric line was damaged underground. The incident did not cause a stoppage in the plant, and did not result in a loss of production. There were no personal injuries.

WHAT WENT WRONG:

- Poor design.
- Lack of attention/care.
- Supervisor failure to identify unsafe condition.
- Inadequate planning/organization.
- Inadequate training/ poor training standards.
- Inadequate work/safety procedures.
- No written job procedure established.
- Rules and instructions not enforced.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Construction work cannot be started without an approved construction plan and, if relevant, an electrical plan.
- During the work permit application, the work permit applicant did not mention and indicate that earthworks would also be carried out. The work permit applicant must have a detailed written task list for the day so that the permit issuer can also assess the risks.
- Communication between the site security company, the construction company, and the project coordination employees needs to be improved. Written records must be made of the meetings and field visits.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

EUROPE ONSHORE

DATE: 21 May 2024
COUNTRY: Romania
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Energy isolation

NARRATIVE:

During rig up operations, the drilling crew was performing mast alignment works. The elevator opened and released uncontrolled the drillpipe joint from around 50cm above the rotary table. The pipe fell on the rotary table tipped over the handrail and came to a rest in the pipe handling equipment on the ground level.

WHAT WENT WRONG:

- Lack of experience to operate the rig.
- No working instruction for rig and equipment.
- Insufficient training.
- Incomplete risk assessment of the drilling rig.
- Design issue for safety system of the elevator and joystick.
- No documentation regarding safety system of elevator.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Additional training with final test to be organize and provided to the rig personnel to operate the rig and equipment, and operating support (advisor/driller coach) must be available for a period time. Translation will be available and course material in Romanian language.
- Inspection and tests of the elevator must be done together with mechanical interlock safety device by the equipment manufacturer. Provide an updated operating manual for the elevator.
- Inspection and tests of the automatic slips must be done by the manufacturer. A report needs to be provided after check. Improve the operating manual to describe the correct removal and installation of the slips.
- Maintenance system must be updated with all the equipment and rig manuals. Daily checks must be performed according with OEM recommendation.
- The rig manufacturer and rig operator should jointly review the Rig Risk Assessment and provide a risk mitigation plan.
- Rig Operator to develop working instruction and procedures for operating and maintaining for this type of rig and auxiliary equipment.
- Reevaluate the rig assessment according with the last version of the rig equipment (for the current rig model and auxiliary equipment).
- Live-Saving Rules dedicated campaign for all rig crews.
- Perform dedicated campaign to develop safety culture.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

EUROPE ONSHORE

DATE: 16 Feb 2024
COUNTRY: Romania
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Safe mechanical lifting

NARRATIVE:

At around 2:45pm, while using a crane to install the access ladder on the working bridge of the installation located at well, an employee lost his balance and fell from a height of approximately 2.5m.

WHAT WENT WRONG:

- Lack of risk awareness for working at height by crew members.
- Improper lifting operations.
- Inefficient location acceptance procedure.
- Inefficient and incomplete training.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Clear establishment of responsibilities during the operations of assembling/disassembling of the sub-structure.
- Identification of risks associated with rig up/down operations.
- Risk assessment for each step mentioned above.
- Perform the Life-Saving Rules modules with the members of the crews (chief of crew to go through the modules with each crew members).
- Compliance with the steps indicated by the manufacturer in the operating manual of the equipment.
- Retraining on Life-Saving Rules.
- Always Stop Work in case of unsafe situations.
- Any violation of Life-Saving Rules automatically leads to the application of motivational management.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment(PPE) not used or used improperly
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision

EUROPE ONSHORE

DATE: 23 Jan 2024
COUNTRY: UK
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

Work commenced on incorrect fan without an isolation in place.

WHAT WENT WRONG:

Permit issued to carry out maintenance on fan belts and grease shaft on a product gas compressor. During work execution it became apparent the work party were working on the wrong product gas compressor.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Procedural non-compliance.
- Regulatory requirements not understood.
- General identification of equipment.
- Inadequate work standard.
- Inadequate supervision.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate supervision

EUROPE OFFSHORE

DATE: 14 Sep 2024
COUNTRY: Netherlands
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Dropped Object – 0.86kg 36mm spanner dropped approximately 25 metres to deck from the exhaust walkway.

WHAT WENT WRONG:

The team had not properly assessed all aspects of the task in respect of the potential for a dropped object from height. Tools were not secured by lanyards and there was no drop prevention matting in place. This allowed the tool to fall through the grating.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- The information recorded on the hazard identification (HIC) must be transferred to the permit to work (PTW) risk assessment.

PTW signatories must properly scrutinise the HIC and associated documents to ensure the hazards and controls are captured on the PTW.

- Ad hoc PTW must be reviewed and authorized to the same standard as planned PTW to ensure they receive the same rigour and scrutiny.

The permit authoriser (PA) and work party must ensure the hazards are well understood and tools/equipment are properly secured to prevent them from becoming a dropped object.

- Duplication of roles such as PA and Area Technician should be held by different people to ensure a level of independent assurance check.
- The prevention of dropped object equipment must be available on the site for use.
- Life Saving Rules must be reviewed and fully complied with.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE OFFSHORE

DATE: 19 Mar 2024
COUNTRY: Netherlands
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

Person observed on top of chemical tank adjacent to handrail.

WHAT WENT WRONG:

A member of the deck crew required access to the filling point located approx. 1.3 metres from the end of the tank to allow them to insert the filling hose, however due to the position of the tank and location of the filling point it was not possible to gain access from the side so the deck crew used a set of step ladders to gain access to the top and rear of the tank and was seen to climb (in a crouching position) onto the top of the tank without the use of fall protection.

- No fall protection was worn.
- The risk assessment and hazard identification (HIC) did not fully cover the task nor working at height aspect.
- Change in operation not identified.
- Risks not fully identified/appreciated.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Lesson Learned issued.

- No fall protection was worn.
- The risk assessment and HIC did not fully cover the task nor working at height aspect.
- Change in operation not identified.
- Risks not fully identified/appreciated.
- Training in working at height was sub optimal.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE OFFSHORE

DATE: 05 Jul 2024
COUNTRY: Norway
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Bypassing safety controls
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

The plan was to move a Drill-Floor-Manipulator-Arm (DFMA) to its parking position. However, the equipment operator accidentally moved the riser catwalk machine instead causing an injury to a person working on the drill floor.

WHAT WENT WRONG:

The wrong type of equipment was moved due to operator error. A person was positioned in the line of fire since there was no plan of moving the catwalk machine.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure the riser catwalk machine is installed with an audible and visual alarm. Reinforce the red zone protocol to prevent the presence of personnel in the line of fire of machinery with potential energy.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 09 Feb 2024
COUNTRY: Norway
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A cutting tool was lifted from the setback area toward the rotary at approximately 0.5 meters height. As the load approached the rotary, it was lowered to approximately 20cm from deck to enable personnel to rotate/adjust the load into the correct angle. The piston rod of the Hydraulic Lifting Tool suddenly disengaged from the cylinder piston. The load dropped to the deck, after which the piston rod leaned and fell sideways, still attached to the top of the cutting tool. At the time of the incident, two persons were in close vicinity.

EUROPE OFFSHORE

WHAT WENT WRONG:

An incorrect/old maintenance manual was referred to in the maintenance system. The daily check list was incomplete. A swivel hook without a swivel bearing had been used which did not effectively prevent the transfer of rotational force from the load to the piston rod. The design of the set screws which should secure the piston rod from coming loose from the piston was inadequate.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure detailed specification of what type of swivel to be used. Update the daily inspection requirements and implement a routine ensuring the latest maintenance manuals are available in the maintenance system at all times.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 25 Jun 2024
COUNTRY: Norway
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

As an Intermediate bulk container (IBC) tank was lifted from the platform deck on a normally unmanned production platform a nearby container containing coiled tubing equipment started skidding. The plate the containers had been stored upon had an overhang below the coiled tubing container. The IBC tank provided the necessary weight preventing the plate from tilting due to the weight of the coiled tubing equipment container. A person working in the area could have been hit by the skidding container.

WHAT WENT WRONG:

The plates that were used as a floor for the storage of containers were not fit for purpose resulting in an overhang. The presence of an overhang had not been properly communicated to personnel involved in lifting of containers.

EUROPE OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure the dimensions of the plates to make up a floor for the containers are fit for purpose (no overhang).
- An installation procedure for correct placement of floor plates to be developed. Follow-up checks to be implemented.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured
PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 29 Aug 2024
COUNTRY: Norway
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

On August 29, 2024, a lifting operation was planned in the shaft at one of our offshore installations according to standard procedures. The task involved transporting a seawater lift pump and associated parts up the shaft. The operation included six deck operators, a logistics discipline lead, a mechanic, a crane operator and the logistics manager, all participating in various phases.

When five pieces of pipe had been lifted out of the shaft with the west crane, a basket with some rigging equipment was lowered down to lift out the seawater lift pump. The basket was guided into a load handling arrangement consisting of rails, which limit the movement horizontally, and three manually-operated hatches. The basket was lowered directly down to the last hatch. This hatch had to be emergency operated to an open position. When the hatch was open, a signal was given to lower the basket to the pump deck. At the same time, two operators moved into the landing area. A nearby deck operator alerted the two operators in the landing area, and they managed to get out of the way before the 2,735 kg basket hit the deck. The team gathered afterwards for a time-out, before the lifting operation was completed.

WHAT WENT WRONG:

The differing situational awareness of the various team members was found to be the direct cause of the incident:

- The deck operator at the hatch thought that the team on the pump deck was waiting for equipment to continue the rigging. The hatch was therefore opened, and a signal was given to lower the basket.
- The crane operator assumed that the landing area on the pump deck was cleared and lowered the basket.
- The deck operator on the pump deck expected to be contacted before the hatch was to be opened. The operators rigging on the pump deck thought that the given hatch was closed and were at times standing in the landing area.

EUROPE OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

A cross learning initiative was implemented at all offshore installations to conduct a one-hour time-out for safety on all shifts based on the established "Safety Alert" created following the incident. Focus on requirements related to blind driving and form of communication during planning and pre-job interviews. Follow up on any findings revealed during this time-out was also part of the initiative.

The incident was investigated internally. Local recommendations were given within the following five areas:

- Fulfilling the role of operational responsible person – lifting operations.
- Training in handling of the hatch arrangement and lifting operations in the shaft.
- Communication within material-handling operations.
- Planning and risk assessment of lifting operations.
- Notification of incidents.
- Clarification of governing documentation (at company level).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 11 Jan 2024
COUNTRY: Norway
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

A gas compression system was shut-down and the pressure in the system was released due to planned upgrade of the control system. Due to a leak in a closed valve between the fuel gas system and the compression system, the compression system was unintentionally re-pressurized.

WHAT WENT WRONG:

The leak in the valve had been known upfront but this degraded barrier had not been properly communicated to personnel involved. The pressure safety valves did not open as intended and control room alarms had not been properly acted upon.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure information of leaking valves are clearly displayed in relevant control room screens. Introduce hydrate prevention of pipes and pressure safety valves.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

EUROPE OFFSHORE

DATE: 03 Mar 2024
COUNTRY: UK
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Energy isolation

NARRATIVE:

When waiting for equipment to be lowered from the top drive to the rig floor during a period of maintenance, a member of the drill crew leant back against the operating lever for the z-back (pipe racking system) turntable at the end of the pipe skate, causing the table to turn and trap their foot. The IP was transferred to the sick bay for assessment and taken to hospital by coastguard helicopter later that evening. Operations were suspended and a safety stand-down was held with the crew.

WHAT WENT WRONG:

When waiting for a piece of equipment to be lowered to the rig floor, a member of the drill crew leant back against the operating lever for the pipe racking system turntable, causing the table to rotate and crush their left foot before being pulled to safety.

The IP was transferred to the sick bay by the emergency response team, assessed by the medic and taken to hospital by coastguard helicopter later that evening and treated for a fracture to the top of the IP's foot.

Incident occurred due to the turntable control being inadvertently activated by the IP with the IP's foot placed in the crush point between hydraulic turntable and beams.

The turntable and control system was tied into rig hydraulic ring line and therefore always live.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Check work equipment on site for:

- Inadvertent operation: Confirm that equipment cannot be remotely or otherwise activated while personnel could be performing normal duties (excluding when locked out/ tagged out for maintenance).
- Safety measures: Check the presence and effectiveness of safety measures, such as guards, emergency stop buttons, warning signs, and protective devices.
- Information and labelling: Ensure that equipment is marked with appropriate warnings and that adequate operating instructions are available and accessible to all users.
- Specific risks: Identify any specific risks associated with the equipment, such as risks from electrical, thermal, or moving parts, and assess the measures in place to mitigate these risks.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

EUROPE OFFSHORE

DATE: 30 Jul 2024
COUNTRY: UK
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During non-invasive temperature monitoring of the west crane engine, as part of repair commissioning, the engineer observed an oil filter leaking (this is a known leak). Whilst investigating the leak the IP placed his hand in an area not protected by the radiator fan guard (engine running) resulting in the hand making contact with the radiator fan resulting in significant injury to fingers of his left-hand and required Medevac to hospital.

WHAT WENT WRONG:

Lack of energy isolation.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Follow the basic safety rule regarding energy isolation.

CAUSAL FACTORS:
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

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DATE: 14 Apr 2024
COUNTRY: UK
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

North crane engine shutdown during supply boat operations this only happens while lifting outboard and down to the vessel with a heavy load (approx 10t).

The engine does re-start straight away and no alarms during incident.

WHAT WENT WRONG:

The exhaust spark arrestors were found to be heavily choked. These were cleaned and replaced but choked up again after a few hours running.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Brand new spark arrestors have now been fitted.

External (contractor) investigation to determine root causes and lessons learned.

EUROPE OFFSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Failure to report/learn from events

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DATE: 11 Apr 2024
COUNTRY: UK
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During inspection activities, at Level 2 Process area, an adjacent flood light failed and fell to deck (4.5m) in the immediate vicinity of the work party (1.5m), striking a valve handle and drains pipework.

Work party observed electrical spark as the cable disconnected.

Work party stopped and cleared the area immediately, contacting the central control room with the site supervisor and REP onsite shortly afterwards.

Circuit was isolated and junction box installed on cable to make safe.

Pipework was inspected with no damage caused other than superficial to the lagging.

WHAT WENT WRONG:

- Floodlight mounting fail resulting in the light dropping to deck (equipment failure).
- Casual Factor: DROPS – Dropped Object Prevention Scheme – no secondary retention or safety securing device and inconsistent approach to DROPS.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Carry out detailed inspection on P1/2 Flood lights across all assets.
- Apply secondary retention.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

EUROPE OFFSHORE

DATE: 28 Feb 2024
COUNTRY: UK
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

South crane boom lower wired handrail failed from corrosion.

WHAT WENT WRONG:

- Handrail failure lies in the absence of a maintenance and inspection regime that had a high probability of mitigating the hidden corrosion damage reaching the point of mechanical failure.
- There were no current maintenance system notifications or work orders driving a planned inspection regime to mitigate the potential of corrosion under sheathed wire handrails.
- The total encasement of steel wires in plastic sheathing when used in the marine environment has significant safety implications, especially when the wire is not regularly inspected and maintained to remain fit for purpose.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Inadequate procurement process and time of installation.
- Inadequate monitoring and assurance of key contractor.
- Inadequate performance management.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 20 Aug 2024
COUNTRY: UK
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Working at height

NARRATIVE:

A team of scaffolders were tasked with the dismantle of scaffold structures erected around chemical tote tanks located in a temporary bund on the upper process deck. An individual sustained a head injury in the area where the dismantling work was being undertaken.

WHAT WENT WRONG:

During scaffold destruct, a standard fell because it was not secured to both ledgers to allow a box-tie to be constructed, as this was determined to be the best method of tie-in due to lack of other available tie-in points. This resulted in the scaffolder being hit by the falling standard.

EUROPE OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Risks associated with using this construction method to be identified ahead of task.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST ONSHORE

DATE: 09 Aug 2024
COUNTRY: Bahrain
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While the rig crew was assembling the Bottom Hole Assembly (BHA), they used a tugger to lift the off-driller side rig tongs to torque the connection. At 1530 hrs, as the tong was being lifted, the winch line pushed the service loop against the transport frame, causing the frame to lift approximately 3 inches and fall onto the rig floor. The transport frame weighed around 200 lbs and was 8 feet tall. No pins were installed to secure the transport frame to the main frame; it was only secured from the top with an L-shaped insert. Fortunately, no personnel were near the transport frame when it fell, resulting in no personal injuries or equipment damage. The facilities were not affected.

WHAT WENT WRONG:

No pins were installed to secure the transport frame to the main frame; it was only secured from the top with an L-shaped insert.

Ineffective communication:

Improper communication between supervisor and crane operator/crane operator was not informed that the webbing belt SWL is 3 Ton, and wasn't informed that the reel brake was ON.

- The pre-job safety meeting was held verbally and did not involve all the parties on the site and was not documented.

Incorrect equipment:

- Contractor was using sling without clear serial number with no tracible certificate.
- The end of the coil string and some areas of the clamp (between the nuts) considered as sharp edges that possibly could cause damage to the sling when pulling the coil.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Final equipment checks:
    - crews must conduct a final check that emphasizes equipment inspections, with specific attention to ensuring all pins are in place for all equipment.
2. Updated inspection checklists:
    - update the daily driller's and floorman's inspection checklists to include checks for the transport frame pin installation.
3. Monthly maintenance:
    - include the monthly maintenance of the transport frame in the enterprise asset management (EAM) system.
    - develop an inspection checklist for mechanics to use during the monthly checks.
4. Standard operating procedure (sop) and job safety analysis (JSA):
    - develop an sop for the installation of the transport frame.
    - create a JSA that includes the pinning of the transport frame and all other required securing pins.

MIDDLE EAST ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 01 Apr 2024
COUNTRY: Bahrain
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A lifting belt broke while pulling the coiled tubing (CT) to be installed into the injector head using a crane. There were no personal injuries, and the facilities are safe.

WHAT WENT WRONG:

The main task was to insert the CT pipe into the Injector Head.

The CT string needed to be pulled approximately 30 feet using an approved third-party rigger and crane vendor.

The crane operator followed the supervisor's instructions, which had been discussed with the CT supervisor.

Proper signals were given, and the desired length of CT pipe was pulled from the reel.

The CT supervisor then instructed the rigger to signal the crane operator to lower the hoist for the next step.

However, the crane operator swung the hoist to the left instead of lowering it, increasing the pulling weight to 6 tons, while the belt's safe working load was 3 tons.

This likely caused the lifting belt to snap, resulting in the CT reel becoming loose and the pipe coming out from the drum.

In addition during investigation the following causes were found:

- Not following procedure/there was no serial number tagged on the webbing belt and no certification for the webbing belt.

Incorrect lifting method/pulling weight increased and reached up to 6 Tons when the belt safe working load (SWL) is 3 Tons, Crane operator did not inform the involved personnel on the site and stop work not applied.

- Damage tools or equipment using uncertified webbing belt.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Inspection and certification of lifting gear:
    - ensure all lifting gear is thoroughly inspected and in good condition.
    - verify that the lifting gear is fit for use and certified, with a visible and clear serial number.

MIDDLE EAST ONSHORE

2. Avoid sharp edges:
   - ensure that no tools being lifted have sharp edges, as these can deteriorate lifting belts.
3. Alternative lifting gear:
   - consider using different types of lifting gear, such as pulling clamps or two-side lifting clamps, to enhance safety and efficiency.
4. Coiled tubing mounting unit:
   - evaluate the feasibility of using a coiled tubing mounting unit to avoid the need for feeding coiled tubing into the injector head.
5. Higher capacity web sling:
   - use a higher capacity web sling (4 tons) instead of a 3-ton sling.
   - apply softeners to cover any sharp edges on the clamp to prevent damage to the sling.
6. Pre-job safety meeting:
   - conduct a toolbox talk (TBT) or pre-job safety (PJS) meeting with the presence of all contractors and company representatives.
   - ensure the meeting is signed and documented for accountability and future reference.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 02 Apr 2024
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Construction
CAUSE: Confined space
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Confined space
SECONARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

Two concrete mixer operators were observed engaging in unsafe practices while working on the Paving work scope. They were found in unauthorized locations, without proper PPE or approval for the Confined Space Entry.

Specifically, one operator was using a jackhammer inside the concrete mixer barrel, a confined space, while the other operator was positioned on top of a ladder platform. No approved documentation or supervision was present, and safe access/egress controls were not in place.

Investigation revealed that the operators were instructed by their supervisor to clean the mixer barrel but were not aware of the CSE requirements or the need for approved documentation. Fortunately, no injuries occurred during the incident.

MIDDLE EAST ONSHORE

WHAT WENT WRONG:

- Unauthorized entry into CSE no approval.
- No approved documentation for task.
- Workers not CSE trained and did not realize the risks involved.
- Lack of supervision.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- All persons to be CSE trained.
- Only enter CSE when trained and all correct documentation and controls implemented.
- Ensure site supervision monitoring CSE.
- No deviation from CSE procedure.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 09 Dec 2024
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Scaffolding team erecting a scaffold platform approximately 8m at height without using correct method of using platform for safe erection. One scaffolder who was positioned not on a working platform and became unbalanced, while holding a scaffold ledger 0.73m length the ledger fell from his hand from approximately 6m from height. The barricaded area was also insufficient with barriers required to extended outwards further so persons are away from potential dropped objects.

WHAT WENT WRONG:

- The worker had the appropriate skills and had been working on the project for more than two years but did not apply those skills to erect the scaffold in a safe manner.
- The scaffolding crew did not provide adequate supervision to ensure the safe control of work, failing to eliminate or reduce risk factors such as the safe methodology for handling dropped objects, WAH, and preventing falls. Additionally, they were under time constraints, which led to shortcut actions and compromised safety.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

On-the-spot practical training sessions have been organized for scaffolders by the scaffolding inspector. These sessions aim to refresh their knowledge and enhance their competency levels, ensuring adherence to safety protocols and proper scaffold erection practices.

MIDDLE EAST ONSHORE

- Supervisors have been instructed to actively monitor scaffolding activities to ensure adherence to safety protocols and proper risk management practices.
- Realistic task schedules are established to prevent time pressures that may lead to shortcut actions.
- All scaffolding activities are closely linked to the Work Permit system, ensuring strict compliance with outlined safety requirements before work commences.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 21 Jan 2024
COUNTRY: Kurdistan Region Of Iraq
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Excavation, trenching, ground disturbance
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Activities to complete the installation of a tie-in pipeline from the metering skid to the existing pipeline involved during the routing of the new pipeline around the periphery of the plant, with multiple underground services and pipelines etc. identified over the route.

The planned activities involved the identification of these buried services/pipelines and subsequent hand excavation in accordance with the Ground Disturbance Certificate (GDC) process. Following exposure of the underground services, mechanical excavation was then utilized to excavate the areas between exposed services (space permitting), down to the specified depth.

On 21st January, an excavator damaged a redundant CCTV cable. Following this event however, an inspection was undertaken by the Senior Inspection Engineer of the nearby 8" gas flowlines and 24" Sales Gas pipeline which identified minor damage.

WHAT WENT WRONG:

- Mechanical excavators operating near exposed critical 'live' hydrocarbon lines.
- Ineffective physical barriers in place to protect critical 'live' hydrocarbon lines with ongoing mechanical excavations occurring nearby.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Specific requirements/limitations on the size of excavators/bucket specifications for use inside production areas.
- Hard stop points to prevent excavators coming within a prescribed distance of exposed hydrocarbon lines (recommend 2 metres minimum).
- Supervision requirements to be defined for any excavations/work ongoing close to exposed critical hydrocarbon/other critical cabling and work to cease if this requirement cannot be met.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate training/competence

MIDDLE EAST ONSHORE

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 24 Feb 2024
COUNTRY: Oman
FUNCTION: Drilling
CAUSE: Exposure noise, chemical, biological, vibration, extreme temperature
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

During well control killing operation by bleeding pressure through choke manifold to flow back tank. Driller smelled gas like rotten eggs in mud tank and choke area. He immediately took a H_2S gas test to check the presence of any H_2S and it was reading more than 10ppm. Immediately, he informed the person in charge (PIC) and they activated H_2S Alarm to evacuate all crew to secondary assembly point, opposite to wind direction. Sent rescue team to check the source of H_2S by wearing SCBA set. Once, he had closed flow back tank the percentage of H_2S was increasing to more than 50ppm. PIC decided to secure flow back tank and removed it out of location.

WHAT WENT WRONG:

- The flow back tank was transferred from another field and not flashed or tested by the 3rd party company.
- 3rd party management inspection not properly done.
- H_2S not checked on the tank prior to start the job.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Need to inspect any unit properly prior to enter the location.
- To get proper document and inspections for any 3rd party unit.
- Flow back tank must be flashed and inspected before sending to any location.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 02 Mar 2024
COUNTRY: Oman
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Bypassing safety controls
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

After loading materials to pickup, forklift operator took a left U-Turn towards CRT tool store, the forklift boom hit crane front driver cabin which resulted in damages on crane driver cabin. At the time of the incident crane operator was in the cabin. Luckily, he was not injured.

MIDDLE EAST ONSHORE

WHAT WENT WRONG:

- Inattention to the surroundings from the forklift operator. Improper positioning of the forklift boom.
- Turning without checking the surroundings.
- Lack of use of vision aids (reverse camera, side mirrors).
- Performing a routine job without being conscious of surrounding hazards.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure to increase attention of the surrounding hazards.
- Always maintain the forklift boom on the downward position while moving.
- Always maintain a clear vision of all surroundings.
- Ensure to identify out of sight hazards while performing any task.
- Ensure to not normalize the risk of any task and stay alert at all times.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 14 Dec 2024
COUNTRY: Oman
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While the pipe spools were being loaded into the HIAB truck using the hydraulic telescopic forklift forks, one of the pipe spools was pushed by the rigger and HIAB operator from the forklift. The pipe dropped onto the HIAB truck and struck a piece of wood, which bounced and hit the rigger's face.

WHAT WENT WRONG:

Investigation not finalized.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Always ensure the team are following Life Saving Rules with particular to line of fire and work authorizations.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

MIDDLE EAST ONSHORE

DATE: 12 Jul 2024
COUNTRY: Oman
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Transport – Land
PRIMARY LIFE-SAVING RULE: Driving
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While employees were swapping feed filter tanks at one of the stations, the old tank was loaded for transport to one of the stores. During transit, as the trailer passed beneath the overhead line (OHL), the vent support hit the fibre optic cable (FOC), causing damage. Unaware of the incident, the driver continued driving and subsequently hit a guard pole rope, detaching one end.

WHAT WENT WRONG:

- Vent support on tank hit the FOC and guard pole rope.
- Guard pole height was 6m and the height of tank vent from land was 6.8m.
- Misjudgement of the height of tank with vent support.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Risk assessment – ensure that potential hazards like collisions with OHL cables or poles during transport for all equipment is included in the risk assessment.
- Planning and communication – ensure that route surveys are part of the execution plan.
- Driver awareness – include OHL and auxiliaries associated risks in the HSE induction.
- Install additional safety barriers – any OHL shall have 2 guard poles (entry/exit), in addition to the required signboards.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 21 Jan 2024
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Safe mechanical lifting

NARRATIVE:

During one of the scaffolding material demobilizations from higher elevation by using handballing methodology, a scaffolding cup lock ledger (9kg) slipped from one of the scaffolders at elevation.

The slipped ledger took a descent and hit a staircase handrailing which is very near and changed it's direction and gradually hit another scaffolder who was among the crew at 4 meter elevated platform.

MIDDLE EAST ONSHORE

He was sent to clinic for first aid and then to hospital.

Scaffold activities by relevant party are suspended and investigation is kicked off.

WHAT WENT WRONG:

- Collision of ledgers due to unexpected change of pace of moving ledgers.
- Deflection of falling ledger by nearby staircase into scaffolder's section.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Communication among work parties during work execution and moments of change.
- Clarity of lead role when handballing.
- Planning of work to mitigate collision and drops during material movement at height. Selection of access method for working at height.
- Managing drop risks during scaffold ledger movement.
- Location of scaffold structure in proximity to structures; protection of workers inside scaffold section.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Work or motion at improper speed
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 13 Mar 2024
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

At 09:30hrs., contractors were manually lifting materials to carry out gas loop blinding for dry gas seal replacement. While lifting the materials using a rope, a hand tool with a mass of approximately 1kg fell approximately 16.6 meters from the tool bag. The falling hand tool hit a scaffolding tube on the ground level that was being used as a barricade and bounced into the leg of a contractor. The contractor was treated with first aid and was able to resume work without any additional medical treatment.

WHAT WENT WRONG:

Personnel hit by dropped object that fell from 16.6m.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- All hazards related to a work activity including those of the environment, such working at an elevated height with the potential for dropped objects should be identified.
- After an incident it is important to communicate all relevant information accurately and promptly.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

MIDDLE EAST ONSHORE

DATE: 23 Aug 2024
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

On 23 Aug 2024 at around 15:30hrs, there was an explosion in the gas chromatograph (GC) lab, which resulted in property damage and burns to a lab analyst. The incident occurred while the lab analyst was calibrating a total sulphur analyzer. During the calibration of the total sulphur analyzer, LPG was released from the cylinder used for the calibration, and the vaporized LPG came into contact with an ignition source causing an explosion and fire.

WHAT WENT WRONG:

Fire and Explosion. IP sustained first and second-degree burns. Equipment damage.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Roles and responsibilities need to be documented and communicated to responsible personnel.
- Complex tasks should be reviewed to identify whether simpler, safer alternatives are available.
- Safety concerns regarding management of change, such as increasing the number of laboratory equipment within the same space, should be evaluated and assessed.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation
PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 30 Mar 2024
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Pin hole leak from 2" methanol injection line at upstream of isolation valve.

WHAT WENT WRONG:

Tier 2 PSE. Significant process safety incident due to gaseous hydrocarbon release 2,592kg (48h) – 1kg/min.

MIDDLE EAST ONSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Thorough Documentation of System Changes

- Material Compatibility with Service Conditions
- Inclusion of All Systems in Maintenance Programs

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 14 Feb 2024
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Work authorization
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Due to suspected water hammering, a rupture occurred on the Low-Pressure (LP) steam header shifted forward while warming up. As a consequence of the line moving forward, the drain point of the LP steam header inside water treatment units impacted the concrete support and was damaged, causing a steam release. Additionally, the force of the impact damaged pipe supports and resulted in some parts of the cladding and insulation being detached and scattered in the area.

WHAT WENT WRONG:

Damage to the drain point, piping support, line cladding and insulation.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Pressurization of the LP steam header should be done properly.
- Awareness of field operators needs to be verified through multiple trainings.
- Encourage reporting incidents and alerting operators.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST ONSHORE

DATE: 05 Mar 2024
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Transport – Land
PRIMARY LIFE-SAVING RULE: Driving
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A bulldozer lost control when coming out from a cement factory community gate and ran over a road crash barrier causing damage to a 4" fuel gas pipeline leading to a desalination plant. This could have resulted in a fire/explosion and serious injuries to people.

WHAT WENT WRONG:

- Pipelines are exposed on the surface and are only buried underground when they cross roads.
- Inadequate barriers to protect the pipelines near road ways.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

A concrete barrier shall be placed opposite to the cement factory community gate.

CAUSAL FACTORS:
PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

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DATE: 20 Feb 2024
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Energy isolation

NARRATIVE:

Finger Crushed in Truck Belt Pulley System. During repair and maintenance, the Injured Person (IP) reinstalled a bracket support for a truck alternator and air conditioner (AC) compressor. After completing the reinstallation, the engine was started to check the alignment of the belt pulley system attached to the alternator and AC compressor. The IP removed his glove from his left hand to check the temperature of the AC compressor by feeling the inlet and discharge pipes with the back of his hand while the engine was running. While doing so, the IP's hand got caught in the belt and was dragged through the pulley, crushing his finger.

WHAT WENT WRONG:

- The IP placed his hand in the Line of Fire.
- The equipment was running while maintenance work was being performed.
- The wrong tool was used for the job.

MIDDLE EAST ONSHORE

What Worked?: The IP was taken to the Medical Aid Centre and then to Hospital for further assessment and medical management.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Always be aware of your hand and body placement while working with rotating equipment and never place your hand in the Line of Fire.

- Do not attempt to conduct maintenance or repairs on equipment while it is running.
- Use the correct tool for the job.
- Be mindful of pinch points and possible hazards or risks that may place you in the Line of Fire.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 25 Sep 2024
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Multiple Finger Injuries: Wind Moves Impeller of Blower Belt and Pulley System.

An Electrical Foreman (Injured Person 1/IP1) was removing a frayed thread from a blower fan belt when a gust of wind moved the impeller, trapping his finger between the pulley and belt. The SHES Manager (IP2) went to evaluate the situation, checked the belt tension in the same place, and also sustained finger injuries. The blower was not connected to any power source at the time.

WHAT WENT WRONG:

The belt and pulley had a guard, but it was insufficient, leading both individuals to place their hands on the belt. They assumed no potential energy was present since the blower was not powered.

- A sudden gust of wind rotated the air blower impeller, which moved the belt and pulley.
- The risk of wind affecting the blower was not anticipated in project procedures, including the Energy Isolation Program (LOTO), Method Statement, and Job Safety Analysis. The equipment manufacturer only advised isolating the power source, which both individuals assumed was sufficient.

What worked?: Case management was followed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Plan for all energy sources, including electric, hydraulic, pneumatic, spring-loaded, gravity, and wind.
- Ensure ZERO energy before starting any work, such as inspection or maintenance.
- Never place body parts into rotating equipment (Line of Fire); ensure guards and signs are in place.

MIDDLE EAST ONSHORE

- Conduct hazard identification and mitigation before entering the incident scene for data collection.
- Discuss solutions with the blower manufacturer to prevent impeller rotation during installation and maintenance.
- Update the energy isolation program and training to include incident findings and emphasise "Lock, Block & Bleed."
- Communicate incident findings to all site personnel during Toolbox Talks.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 10 Feb 2024
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Transport – Land
PRIMARY LIFE-SAVING RULE: Driving

NARRATIVE:

Trailer tub tip over incident resulting in abrasions on driver legs and trailer tub damage.

WHAT WENT WRONG:

While travelling on the road outside the fence the driver, in a state of drowsiness, veered off approximately 4 metres from the road. Upon realising this, the driver took a sudden left turn in an attempt to return to the road. However, this caused the trailer tub and the head to become misaligned. All the resulting force was exerted onto the connecting bolts, causing all of them (8 bolts) to break. This resulted in the trailer tub tipping over onto the driver's right side without impacting the trailer head. Due to the sudden strong vibration in the cabin during the incident, the Injured Person (IP, the driver) sustained minor abrasions on both knees and on his left leg. At the time of incident the IP was wearing his seat belt.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Provide training to drivers on how to identify fatigue and react in such scenarios.
- Drivers shall be made aware of the importance of having enough sleep to avoid drowsiness while driving.
- Drivers must be given the flexibility to park their vehicles in a safe place and take rest if they feel sleepy while at work.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

MIDDLE EAST ONSHORE

DATE: 17 Jan 2024
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Transport – Land
PRIMARY LIFE-SAVING RULE: Driving
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A Contractor driver, while reversing a 1 ton pickup, hit an Instrument Foreman crossing at the back of vehicle. IP fell and was run over by the rear tyres of the vehicle, his final position after the impact was face up, under the vehicle, between the front and rear tyres. He sustained multiple injuries (fractured right ribs, abrasion to face, cuts, and burn to right hand).

WHAT WENT WRONG:

- Inadequate road safety risk management process in the area.
- Lack of established systems for monitoring driving behaviours and ingrained road safety culture.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Develop and implement a traffic management plan for the area considering site-specific needs to minimize the risk of accidents.
- Implement road safety procedures to facilitate the efficient movement of man, materials and equipment.
- Implement effective system for monitoring safe driving behaviours to encourage safe driving practices.
- Conduct regular inspections and audits of road safety procedures to identify potential hazards or areas for improvement.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

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DATE: 16 Apr 2024
COUNTRY: Qatar
FUNCTION: Unspecified
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

An overhead crane fell on a portacabin and pipelines. During inclement weather conditions with storms and high winds, an electric overhead traveling (EOT) crane collapsed and fell onto an adjacent portacabin and molten sulphur and heavy fuel oil pipelines without any personal injuries.

MIDDLE EAST ONSHORE

WHAT WENT WRONG:

- Strong winds forced the unsecured EOT crane to roll to soft ground and topple.
- Crane operators were not oriented on applying anti storm lock (ASL) while gantry crane is in parking position or not in use.
- Fault in ASL identified during maintenance/certification of crane, but no action taken.
- Portacabin last minute addition to project scope without proper risk assessment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Apply physical locks/securing devices when required. Anti storm lock shall always be applied when crane is not in use.
- Familiarize workers on specific equipment features, like safety critical devices, to ensure safety of that equipment/asset while in use or idle.
- Any defects/malfunctions identified during maintenance or inspection, to be properly communicated and addressed to ensure safe operations.
- Last-minute scope addition to management of change, shall be thoroughly reviewed prior to executing the additional scope of work.
- All Asset Owners to carry out a review. Ensure any portacabin or pipelines are placed after assessing any potential line of fire situation.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

MIDDLE EAST OFFSHORE

DATE: 16 Apr 2024
COUNTRY: Qatar
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During high winds of about 50 knots, one of the sections of the top deck muster station sunshade weighing about 39.15kg was dislodged, blew off, and fell about 70 feet down onto the cantilever deck which was in tow position in preparation for planned rig move. No personnel injury because all personnel had earlier been asked to remain indoors due to the predicted bad weather. No other property damage from the incident.

WHAT WENT WRONG:

- Poor design. The design focused on its primary purpose (sun-shade) and allowing easy removal to spot equipment in the area beneath the shade. This led to the decision to minimize restraining devices in order to make removal less difficult.
- Poor risk perception: During the design stage, extreme high winds conditions, which could cause a blow-off effect on the sunshade securing mechanism was not taken into consideration.
- Poor management of change and oversight: The addition of the change was viewed as minor. Consequently, no formal design process, with external review and internal management approval was adopted.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Cover the sunshade installation, including addition of ratchet straps, under routine inspection regime. To be added into DROPS inspection schedule.
- Conduct a hazard hunt on the rig to identify if any other structures have been installed that have not undergone formal review and approval and therefore pose potential threats.
- Use the MOC to formally assess the robustness and adequateness of the ratchet straps and securing mechanism until the external design process is complete.
- Develop a Fleet-wide checklist for Adverse Weather Conditions.
- Review internal management system processes and demonstrate how similar engineering works should be managed via the MoC process.
- Design documentation shall include risk assessment and be approved at appropriate levels.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

MIDDLE EAST OFFSHORE

DATE: 13 Apr 2024
COUNTRY: Qatar
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Hybrid bridge plug dropped onto drill floor while lifting; during lifting operations, hybrid bridge plug was lifted horizontally from cantilever to the drill floor by port aft crane. Top Drive System (TDS) elevators were latched onto the top of hybrid bridge plug and lifted to vertical position in order to disconnect crane wire sling. As soon as hybrid bridge plug was close to vertical position, the running tool released causing the hybrid bridge plug to drop on the rig floor. Weight approx. 850 Kg – length 27 ft – dropped from around 33ft.

WHAT WENT WRONG:

Improper make-up/assembly:

- Running tool and running tool sub were misaligned while installing the shear screws.
- Shear screws did not bottom as required inside each shear screw holes.
- The running tool can be stab in the running tool sub in three different positions, but only one position allows the shear screws to bottom in the respective holes.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Update the assembly preparation procedure and onshore inspection check list.
- Implement a dual check process.
- Update hybrid and inflatable assemblies' installation procedure.
- Redesign the running tool.
- Create an offshore checklist prior check of activity.
- Red zone management to be reinforced and update job safety analysis.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

MIDDLE EAST OFFSHORE

DATE: 13 Jun 2024
COUNTRY: Qatar
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

At the time of the event, team were performing a static well logging survey prior to drilling. During the period of stationary block activity, four crew members were approved to enter the rig floor red zone to install the mousehole prior to drill ahead. While members located at the rig floor near the v-door, awaiting receipt of the mousehole from the port aft crane, a half-moon segment, weighing 3.2 kg, fell and landed approximately 1m away from the personnel.

WHAT WENT WRONG:

- The four segments of the lifting clamp have not been removed completely.
- Proper secondary retention on back up kelly hose had not been installed at the first rig up.
- Corrective action preventive action was ambiguous, and the observations made during hazard hunt were not considered and pictures were not linked.
- Design of lifting clamp is not self-safe while removing (optical illusion).
- Lack of supervision: no permit to work, no job safety analysis and the instruction were unclear.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Review the design of lifting clamp.
- Corrective action preventive action system to be upgraded to link observation and picture together.
- Culture model to be initiated to implement personal procedural compliance realignment.
- Job safety analysis for changing lifting clamp to be updated (potential lifting clamp made-up of 4 elements to be mentioned).

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

MIDDLE EAST OFFSHORE

DATE: 26 Jan 2024
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Transport – Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Bypassing safety controls
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A vessel lost control, went under a production station South side and collided with pipelines breaking instrument airline, nitrogen line, diesel bunkering line, and water line. As a result, the production station was shut down for some time. This is an incident that could have resulted in multiple fatalities.

WHAT WENT WRONG:

- The sea current, swells, and wind pushed the vessel towards the platform.
- Poor preparation, planning, and execution of the personnel transfer.
- The vessel manoeuvre protocol at offshore structures was not followed.
- Master was not at the controls of the vessel.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Prior to approaching the platform, the vessel's crew should thoroughly plan the manoeuvre, assessing weather conditions, tide, and current patterns.
- The Master should be at the controls, with the Chief Officer observing the vessel hull, providing clear instructions on the distances and vessel movements during the manoeuvre.
- Pre-departure checks to be conducted prior to departure, with roles and responsibilities clearly defined.
- A person from the manned platform to be present to support mooring and unmooring operations.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 28 Jun 2024
COUNTRY: Qatar
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Electro-chlorinator unit cell cover blown off causing multiple injuries. Following a corrective maintenance activity carried out during the day, the Programmable Logic Controller (PLC) of the Electro-Chlorination Unit was reset. After the reset, a violent explosion occurred in the Electro-Chlorination Unit, causing a blast and multiple injuries to the Injured Person (IP) who was in the immediate vicinity. The general platform alarm was initiated, and emergency

MIDDLE EAST OFFSHORE

protocols were followed. The IP was found lying on the deck. He was transferred to the clinic, and then safely evacuated to a Doha hospital by Medevac chopper.

WHAT WENT WRONG:

- Respect Hazards: Hydrogen detection was not adequately addressed in the current design.
- Apply Procedures: Operating and maintenance practices did not follow the
- recommendations from the Original Equipment Manufacturer (OEM). There were gaps in the application of Company procedures (e.g., Permit to Work).
- Watch for Weak Signals: There was a failure to apply executive action in response to the observed degradation of various systems (Management of Change, quality control, surveillance, emergency management, and the Permit to Work system).
- Maintain Competency: There was an absence of Hydrogen risk awareness among offshore operations personnel.
- Manage Change: No Risk Assessment or HAZID/HAZOP was carried out within the framework of the eMOC, and a last-minute change was implemented without following the eMOC procedure.

What went well?: Emergency Response was immediately initiated, and Case Management was followed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure that all steps of the MOC process are effectively applied and that all expected deliverables are issued at the required level of quality.
- Verify that any Hydrogen risk is adequately addressed in the safety reviews of Electro-Chlorination units, including the limitation of personnel exposure.
- Verify the adequacy of the Equipment Strategy with the vendor's recommendations.
- Ensure that executive action is taken when degradation in site conditions, operating parameters, or procedures is observed.

Provide Hydrogen risk awareness sessions and training to all personnel exposed to Hydrogen risks.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

MIDDLE EAST OFFSHORE

DATE: 30 Jan 2024
COUNTRY: Qatar
FUNCTION: Construction
CAUSE: Struck by (not dropped object)
ACTIVITY: Transport – Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Bypassing safety controls
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Dive support vessel contacted with non-commissioned platform: the saturation diving vessel was positioned 750 meters southeast of the affected platform, awaiting favourable weather conditions. The vessel's dynamic positioning (DP) system was engaged, with yaw under automatic control and surge and sway managed via joystick control. The senior dynamic positioning operator made a surge adjustment on the joystick to counteract the wind and sea forces acting on the vessel. Subsequently, the operator commenced reviewing some documents and failed to recognize that the vessel had entered the safety zone of platform and increased its forward speed to 1.5 knots, resulting in contact with platform.

WHAT WENT WRONG:

The company requirements were not adhered to at the vessel's bridge. Additionally, there was a failure in fundamental navigational seamanship to safely position the vessel during extended standby periods. Inadequate risk perception and situational awareness.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Enhance the technical capabilities of the Vessel Monitoring and Alert System (VMAS).

Additionally, to update the company procedures to include:
- Provide guidance on vessel position keeping during standby periods.
- Identify VMAS performance expectations, roles, and responsibilities.
- Identification, Definition, and Organization of Offshore Sites: Describe the process for updating the field for current and future VMAS, Geoinformatics team, etc.
- Create a new document defining criteria and requirements for Company Representatives, Marine Representatives, and other roles on the vessels, including onboarding processes, formal appointments, and technical and HSE reporting lines.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication

NORTH AMERICA ONSHORE

DATE: 16 Jun 2024
COUNTRY: Canada
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While laying out 114.3mm tubing, the injured person (IP) was assisting the floor hand with the tubing tongs and clamps. As the floor hand hoisted the tubing tongs to install them on the stump, the IP grabbed the handle of the open tong door latch to guide the tongs into place, but the tongs were not properly aligned with the stump. When the tongs began to move forward, the IP's fingers were caught between the handle and the stump, resulting in multiple fingertip amputations on their right hand.

WHAT WENT WRONG:

The injured worker grabbed the handle of the open tong door to guide the tongs into place, resulting in their misalignment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Handles on tong doors need guarding to protect workers from impact. Job Safety Analysis (JSA) and job procedures must clearly specify the correct use of equipment handles to prevent misuse. Reducing human-equipment interfaces through the use of remote controls and automation.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 22 Oct 2024
COUNTRY: Canada
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

Two workers were attempting to install a 42" steel beam weighing approximately 100 lbs. During the process, they accidentally knocked the beam out of position. It fell approximately 5.5 feet onto the scaffold deck below, damaging the deck. The beam then deflected off the deck and fell an additional 5.5 feet to the grating below. No injuries occurred during the event.

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WHAT WENT WRONG:

The workers installing the beam were inexperienced with the task, and had no supervision/guidance as the general foreman took the day off. Appropriate barricades and exclusion zones were not established by the workers.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Adequate supervision and guidance are essential, especially when workers are not fully competent for the task. Tasks should only be assigned to workers who are competent and have been evaluated for the required competencies. Field Level Hazard Assessments (FLHAs) must be thorough and complete to identify and mitigate potential hazards.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 09 Jul 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During night operations, a worker was pulled into the drill pipe causing him to strike his face and fall. He received facial lacerations and lost a front tooth.

WHAT WENT WRONG:

During the removal of the rotating head for trip nipple installation, it was pulled through the rotary table. While connecting the tugger, the worker positioned himself within the slack of the tugger line. When the rotating head slipped past the tool joint, the slack was taken up and trapped the worker under his left arm and pulled him into the drill string.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Job tasks need to be completed by the designated team members from start to finish.
- Be mindful of body positioning and watch out for others – keep yourself and others out of the line of fire.
- Explore options on how to safely assist pulling/handling the rotating head and revise the process.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Work Place Hazards: Congestion, clutter or restricted motion
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 14 Mar 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A forklift was tipped on its side after attempting to unload a high-pressure hose off the back of a frac pump. The forklift operator was lifting one end of the hose while the pump was pulled forward. The end of the hose that was laying on the back of the pump was caught on the hose rack and pulled the forklift with it. No injuries were reported.

WHAT WENT WRONG:

The forklift operator signed out the forklift identifying a spotter that did not participate in the task.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The contract company did not have a standalone JSA available for personnel to use when non-routine operations became present. The current JSA for forklift operations is pre-filled and not sufficient to address changing job conditions.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

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DATE: 28 Feb 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A 43 foot long assembly of downhole tools and components fell 3 feet to the ground when the wireline cable parted. The assembly was suspended in a vertical position and when the lubricator was lifted, it impacted the wireline sheave, and the wireline cable parted. This caused the assembly to lay over and come to rest partially on the rig floor. There were no personnel on the rig floor at the time of the incident. No injuries were reported.

WHAT WENT WRONG:

Inattention and lack of situational awareness.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Roll out JSA Management Training with HSE included. This will be done on location with rig crews and other third parties.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

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DATE: 08 Nov 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Safe mechanical lifting

NARRATIVE:

A misalignment of the pin to the box when stabbing a stand to add to the drill string allowed the stand to rotate/roll outward and off of the stump. The elevators had been lowered enough to prevent them from stopping the stand from falling. The stand fell approximately 3' and simultaneously struck IP's right foot and the rig floor.

WHAT WENT WRONG:

Failure to mitigate hazard – reported ongoing use of a malfunctioning stabbing guide.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Ensure that a functioning back-up stabbing guide is on-site.
- Ensure correct pin-to-box positioning before lowering TD/elevators.

CAUSAL FACTORS:
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 06 Feb 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While holding a flowline aloft with a forklift in an attempt to clean it, the nylon strap used to suspend the flowline broke and allowed it to drop onto the cab of the forklift.

WHAT WENT WRONG:

Improper lifting techniques.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Ensure proper lifting devices are used when performing lifting operations.

CAUSAL FACTORS:
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 05 Apr 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A wellhead spool and stinger was dropped from the top of the wellhead due to improper rigging. A two-leg chain listing sling that was missing the safety hook latches were used to lift. One of the hooks came loose as the crane lowered the stinger into position causing the stinger to fall to the ground. No injuries were reported.

WHAT WENT WRONG:

- Inadequate Practice: Faulty rigging and hoisting equipment was used.
- Field personnel making the up lifting and rigging equipment had not been trained properly.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- HSE to conduct site audits focusing on lifting and rigging operations.
- Training specific to proper rigging practices to be conducted by vendor for applicable parties.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

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DATE: 23 Aug 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Bypassing safety controls
SECONARY LIFE-SAVING RULE: Safe mechanical lifting

NARRATIVE:

A contract workover crew was in the process of pulling an electrical submersible pump (ESP) when the traveling block contacted the crown of the derrick. The rig is equipped with a crown and floor protection system, but it was not in operation at the time of incident. The crown and tubing line were inspected by a rig manufacturer representative and no damages were found. Operations resumed later the same day, and the floor protection system was returned to service.

WHAT WENT WRONG:

The contract crew disabled the floor protection system due to not being able to set correctly.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

All contract workover rigs were made aware of incident during a safety standdown, and re-training/assurance of properly functioning floor protection system was mandated.

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CAUSAL FACTORS:
PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

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DATE: 28 Aug 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Bypassing safety controls
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A high-pressure discharge hose on a frac pump failed mid stage.

WHAT WENT WRONG:

Valve was actuated, opening flow to the unprimed pump, causing an overpressure event in the hose and parting the hose at the crimp closest to the fluid end.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Reinforce procedural compliance and training to contract crews.

CAUSAL FACTORS:
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

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DATE: 24 Jan 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Energy isolation

NARRATIVE:

A C-clamp on a high-pressure hose came apart at the hub while pumping 42 BPM at approximately 8,900psi. The line restraint was ripped in half and the hose whipped causing damage to the hot swap pod and a compressed natural gas line that was laying alongside of the frac pump. Near miss.

WHAT WENT WRONG:

Bolt up and torque procedures were inadequate leading to a failure during pumping operations.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Correct procedure, identify a visual confirmation method to ensure bolt is proper tightness to the correct specs.
- Document proper torque spec confirms delivery of torque tool.

CAUSAL FACTORS:
PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

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DATE: 08 Jul 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Safe mechanical lifting

NARRATIVE:

The Driller was lowering a stand with the top drive after the rig service. At the same time, the floor hands were putting another stand in the mouse hole. While the IP was attempting to insert the stand in the mouse hole, it stacked on the rig floor. Then, the top drive that was lowering hit the stacked-out drill pipe causing it to bow. When the bowed drill pipe released energy, it swung away initially and then back towards the IP and impacted him. This resulted in chipped teeth and a laceration to the IP's eyebrow.

WHAT WENT WRONG:

N/A

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

N/A

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 07 Jul 2024
COUNTRY: USA
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Safe mechanical lifting

NARRATIVE:

A pulling unit's blocks dropped due to a pin coming out of the brake linkage system while running in the hole with stands of tubing.

WHAT WENT WRONG:

- Human machine interface – obstructed view and in a cramped position, worker has a hard time seeing item: due to the position of the linkage pin within the draw works' machine guarding, it was difficult to visually verify cotter pins were installed.
- Human performance – standards, policies, and administrative controls (SPAC): no post brake job checklist in place.
- Human performance – quality control needs improvement: contractor daily inspection did not specify which critical parts to verify visually.
- Contributing factor: Crown out floor out controller (COFO) was not setup to the disc assist in addition to the main brake.

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CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Retro fit all rigs with upgraded pin design/concept that makes inspections easier to get right, and harder to miss. Also cut windows/doors where able, to make hard to inspect items easier to see.
- Implement an inspection checklist that is performed by rig leadership after any work is performed on the braking system.
- Contractors daily startup inspection will be updated to identify "must observes" specific to the drum brake inspection.
- Required to tie in the COFO to disc assist (if present) in addition to the main brake as a best practice.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Work Place Hazards: Congestion, clutter or restricted motion
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 18 Sep 2024
COUNTRY: USA
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Rig was conducting Drill Out Operations when centrifuge had a catastrophic failure with the motor and transaxle being separated from the rest of the equipment. Nobody was hurt.

WHAT WENT WRONG:

- Equipment difficulty – no preventative maintenance (PM) for equipment.
- Human performance – training – triggers not recognized as malfunction.
- Equipment difficulty – specifications need improvement – debris not contained by equipment design.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implement a 3 step process to secure all centrifuges:
  - Verifying all bolts are torque to spec.
  - Installing engineered restraining band.
  - Implement PM program.
- Service provider to develop and implement a maintenance tracking/planning and preventive maintenance system.
- Operator to evaluate service providers' quality programs during selection.
- Service provider to analyze equipment malfunction scenarios to create standard procedures for their operators.
- Service provider to train and validate knowledge of their operators on malfunction diagnosis and remedy.
- Operator to evaluate requirement for retaining guards on centrifuges.
- Service provider to evaluate adding engineered retaining guards.

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CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Organizational: Inadequate training/competence

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DATE: 15 Apr 2024
COUNTRY: USA
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Operator was diagnosing a 'no flame detected' issue on the reboiler. After manually purging, the unit operator initiated the lighting sequence. There was residual gas in the burner tube that ignited, causing a loud popping sound and causing the bird cage to come off the burner stack.

WHAT WENT WRONG:

N/A

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

N/A

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 10 Apr 2024
COUNTRY: USA
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

At approximately 10:30pm on 04/10/2024, the field monitoring room (FMR) contacted on-site flowback personnel notifying them of a bradenhead increase. A company production employee arrived on location to visually inspect the transducer which showed 138 psi. After they confirmed pressure, the FMR instructed them to bleed off pressure through flowback package. Upon finishing the stage, zero pressure was confirmed and then bleed off operations commenced. Contractor employee verified valve alignment in the vessel and isolated. During walk through, Contractor employee noted pressure on the inlet line to the vessel showing approximately 600 psi. They proceeded to ignite the combustor and bleed off pressure into the vessel while controlling the flow by choke. They then verified

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that they were getting gas in the combustor and seeing a flame. After burning for approximately 1-2 minutes the flowback tank to flash back and ruptured causing damage to the tank. No LOC reported. No injury reported. Stop work was initiated and operations was halted. No other damage reported.

WHAT WENT WRONG:

N/A

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

N/A

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 11 Feb 2024
COUNTRY: USA
FUNCTION: Production
CAUSE: Exposure electrical
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Work authorization
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While troubleshooting a switchgear power distribution cabinet, a technician's voltmeter malfunctioned due to excessive voltage. The circuit's voltage exceeded the meter's internal resistance, causing it to produce a loud noise and the display module to detach from the unit. The unit produced an electrical charge that cause visible damage to the test lead and test probe. The electrical gloves the electrician was wearing received black electrical marks.

WHAT WENT WRONG:

The technician was working on equipment that he was not familiar with or qualified to work on. The technician thought he was testing low voltage equipment and failed to recognize the voltage levels in the cabinet. The contact with the medium voltage circuit resulted in the overloading the meter.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- High voltage labels need to be clear and conspicuous.
- Facility orientation including electrical/instrumentation classification needs improvement.
- A better understanding of the arc flash requirements across the Midstream asset is needed.
- Training and compliance on the electrical safety program needs improvement.
- Upstream technicians should work on Midstream facilities unless they have been trained and orientated on a site-specific basis.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

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PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 02 Jul 2024
COUNTRY: USA
FUNCTION: Production
CAUSE: Exposure electrical
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Energy isolation
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A contract line crew was attempting to replace the crossarm of an energized three-phase overhead power line (14,400 Volts) when the wrap-lock parted on the centre phase insulator. The energized centre phase contacted the ground wire on the electrical pole causing a phase-to-ground fault which resulted in an arc. The arc caused burn marks on the electrician's right rubber glove and the bucket he was working in. The line was deenergized and repairs were made. No injuries were reported.

WHAT WENT WRONG:

Failure to follow procedure with regards to safety protocols required for energized cross-arm changeout.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Contract work crew re-training on energized work procedures. Reinforce use of PTW process.

CAUSAL FACTORS:
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

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DATE: 20 Sep 2024
COUNTRY: USA
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

On September 20, 2024, at a frac pond, while a new transformer on a utility pole was being commissioned/energized by a contractor employee qualified electrical worker (QEW), the transformer faulted internally causing its lid to propel off and land on the ground away from the pole. Following the work practice, the QEW had positioned himself outside of the debris field and company defined safe area of 10 feet from the base of the pole, and all other workers were removed from the area during energization. The transformer lid, weighing ~23 lbs, fell from a height of 21 feet, and landed 11 feet from the base of the pole. The QEW had positioned himself 12 feet from the base of the pole and the lid landed 6 feet from the QEW.

NORTH AMERICA ONSHORE

WHAT WENT WRONG:

N/A

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

N/A

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 04 Apr 2024
COUNTRY: USA
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Excavation, trenching, ground disturbance
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Workers were setting utility pole when control of pole was lost resulting utility pole falling on cab of an adjacent truck. No personnel were in the cab of the truck.

WHAT WENT WRONG:

- Known production bias – contractor works for other operators under different process guidance.
- Trigger not recognized on use of choker to raise pole with habit of sliding down when in use.
- No recognition of simultaneous operations (SIMOPS) during 2 work crews in close proximity.
- Error Prone situation with boom truck operator controls.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Field leadership to understand short service worker programs of the contractor companies supporting their team and ensure they are followed.
- Company supervision must be present when excavation work is performed.
- Technical superintendent to engage with contractors to enhance identification of equipment control panels.
- Superintendents to ensure and address outside incentives and the expectations of completing work without production bias while working.
- Superintendents to ensure effective work authorization discussions and safe work checks with contract service providers.
- Superintendents to ensure for SIMOP discussions within same operations are conducted.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Work or motion at improper speed
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

NORTH AMERICA OFFSHORE

DATE: 28 Nov 2024
COUNTRY: Mexico
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During singling in hole 5" drill pipe, the Driller retracted the top drive system (TDS) dolly at rig floor level and observed a dropped steel plate from the TDS to the port side of the drill floor. Operations suspended, well secured, carry out DROPS inspection and observed that the roller bracket mounting plate had sheared in several spots. Investigation ongoing.

WHAT WENT WRONG:

IMMEDIATE CAUSES:

Substandard conditions – Defective tool/equipment: design failure/insufficient. Web Roller mounting cracket inadequate to sustain forces applied during operations.

BASIC CAUSES:

Inadequate maintenance/inspection – inadequate communication of corrective maintenance needs: conduct a thorough review of all product bulletins issued for critical equipment and updated preventative maintenance system as needed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

CONTROL AREAS FOR IMPROVEMENT ACTIONS:

Near miss and substandard conditions – drilling contractor has not been able to swiftly and expeditiously address management system shortcomings to prevent reoccurrence of similar events.

CAUSAL FACTORS:
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate communication

NORTH AMERICA OFFSHORE

DATE: 16 Oct 2024
COUNTRY: Mexico
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While tripping in the hole and whilst making a connection with the iron roughneck the upper clamp jaw spring assembly retention bolt was sheared and as a result it was propelled approximately 2.5m from its initial location on the iron roughneck and came into contact with the doghouse glass window pane causing it to shatter. This resulted in reduced visibility through the window.

At the time of this event, no personnel were present within the red zone and therefore nobody was in the line of fire. Nobody was hurt.

Operations were immediately stopped and investigation commence.

WHAT WENT WRONG:

IMMEDIATE CAUSE

Substandard Conditions – Inadequate Guards and Barriers: wear and improper alignment of tool can cause bolt(s) to be sheared.

BASIC CAUSE

- Inadequate Maintenance and Inspection
- Inadequate Communication of Corrective Maintenance Requirements: during rig acquisition, rig intake and routine inspection process, PIB's issued in the past for this equipment weren't taken into consideration.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Controls areas for improvement actions:
- Asset Integrity Review – rig recommission process to reach 'ready for operations' failed to be exhaustive enough to ensure asset reliability.
- General Conditions Check – preventative maintenance systems failed to incorporate full equipment history within system.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Organizational: Inadequate communication

NORTH AMERICA OFFSHORE

DATE: 26 Dec 2024
COUNTRY: Mexico
FUNCTION: Drilling
CAUSE: Unspecified – Other
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A 5 7/8" drill pipe came off the catwalk and came to rest on the cantilever deck. There were no personnel at risk during the incident.

At 14:15, while laying out joint #63 of 5 7/8" drill pipe onto the catwalk machine, the joint came off the catwalk conveyor.

After opening the elevators and raising the catwalk arm, the joint bounced off the catwalk machine, slid down the side of the V-door, and came to rest on the cantilever deck. The No-Go zones on the catwalk, rig floor, and cantilever were clear of personnel during the incident.

WHAT WENT WRONG:

IMMEDIATE CAUSES:

Substandard acts and practises – improper operation of tool/equipment/machinery/device: inadequate communication and coordination between the driller and the assistant driller, led to improper operation of top drive system (TDS) and pipe transfer conveyor.

BASIC CAUSES:

Job/system factors – inadequate work/production standards – inadequate risk identification/evaluation in development of standard: contractor work instruction/risk assessment was not relevant to task / was not specific to task.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Control areas for improvement actions:

Planning and Administration – Work Planning and Control: The Work Instruction/Risk Assessment being used was not relevant to the task. Failure at a supervisory level to ensure all relevant documentation and procedures were in place.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication

NORTH AMERICA OFFSHORE

DATE: 19 Mar 2024
COUNTRY: USA
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Electrical workers had completed work in the crown and were preparing to return from level 4 (144' level) in the derrick to the drill floor level via the derrick elevator. As the inner derrick elevator doors opened, the workers heard a noise and observed the lower section of the elevator door become dislodged and descend 43 ft to the derrick level 3 (101' level). No personnel were in the derrick level where the dropped object occurred.

WHAT WENT WRONG:

The securing rivets for the lower section of the elevator door failed allowing the lower door section to come free from the connecting rails on the side of the door/upper trim and drop down to the next derrick level and come to rest outside the level 3 elevator access. It is believed that the securing rivets did not all fail at the time of the event but over time resulting in a degraded securing capability.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Elevator door needs a sign to state: "Do not use force when operating door".
- Guidance on lifespan and/or recommended change out of rivets needed.
- Design secondary retention (extension of door) to retain the door to prevent a drops hazard.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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DATE: 08 Feb 2024
COUNTRY: USA
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

During the removal of scaffold materials from the helideck to the main deck, a worker inadvertently dropped an 18 pound scaffold pole 28 ft. to the main deck below. It landed in an unguarded area of the main deck walkway. No one was injured.

WHAT WENT WRONG:

A worker briefly rested an 8-foot pole on a handrail while waiting to lower it. When he lifted it, the pole caught the end of the handrail and pulled free from his grip. The pole then fell through a gap in the handrail.

NORTH AMERICA OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Incorrect body positioning led to the worker's inability to control the pole.
- A secondary dropped object review was not performed, as it was not in the company's current Dropped Object Prevention Procedure.
- Because the work zone was defined near the living quarters, the staging and unloading area's high risk for dropped objects was overlooked.
- No pre-job walkdown by the Permit Issuer led to the issuer missing identifying that this area was a hazard.
- The Permit Issuer failed to identify the hazard because they did not conduct a pre-job walkdown

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Overexertion or improper position/posture for task
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 06 Sep 2024
COUNTRY: USA
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Bypassing safety controls
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

An employee (referred to as the Injured Party or IP) experienced a fall from the second stair while descending a stairwell. The incident occurred due to the IP tripping over a piece of rope that was obstructing the stairs.

WHAT WENT WRONG:

- Materials were improperly stored at the top of the stairwell, including a bundle of loose rope that protruded into the path, creating a tripping hazard. This represents a deviation from site standards.
- The IP acknowledged a lapse in awareness and recognized their complacency while descending the stairwell.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Continue to conduct weekly Hazard Hunts and establish clear expectations with teams regarding the maintenance of housekeeping standards in the vicinity of access and egress routes, as well as stairwells.
- Enhance personnel's situational awareness aboard by continued structured weekly observation trends, conducting team leader assessments, and scheduling leadership site visits.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress
PROCESS (CONDITIONS): Work Place Hazards: Congestion, clutter or restricted motion
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 14 May 2024
COUNTRY: Azerbaijan
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the unloading of the cementing unit from the truck using 2 mobile cranes at the cementing site, due to improper rigging the bulky load became unstable. A worker tried to walk under the suspended load but was stopped by signalman.

WHAT WENT WRONG:

- Inadequate work planning.
- Lack of supervision.
- Lack of competency.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Competency training program to be enforced.
- Awareness sessions to be conducted.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 09 Feb 2024
COUNTRY: Azerbaijan
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the lifting and dismantling of the tool at the well drilling site, a 178 mm conductor was not properly attached to the nylon rope and fell from a height of approximately 0.8 m onto the driller's assistant standing below. The worker was pulled aside and was not injured.

RUSSIA & CENTRAL ASIA ONSHORE

WHAT WENT WRONG:

- Inadequate risk assessment.
- Inadequate control of work procedure.
- Lack of supervision.
- Lack of competency.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Control of work procedure to be enforced.
- Competency training program to be enforced.
- Risk awareness sessions to be conducted.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 17 Jan 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Transport – Land
PRIMARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

Derailment incident happened after butane discharge operation, one of the empty rail tank car derailed on a curve. The car was put back on the track. As a result, there was no damage to the equipment, the environment, or people.

WHAT WENT WRONG:

- Inadequate work planning.
- Inadequate site inspection.
- Lack of supervision.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

The movement of trains through entrances and exits of rail sidings should be supervised by a competent person.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 23 May 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the replacement of the stone cladding of the facade of the plant administrative building by the contractor, the protective net stretched over the scaffold erected over the entrance to the canteen was torn, posing a danger to the workers entering the canteen. During the stone cutting, stone chips and dust were observed falling to the ground. No injury was sustained.

WHAT WENT WRONG:

- Inadequate contractor management procedure.
- Lack of supervision.
- Poor safety culture.
- Inadequate risk management.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Contractor working on site should be monitored more closely.
- Awareness session to be arranged involving contractor workers.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 01 Jun 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A 15-meter-high lighting pole fell onto a fence of the plant due to strong winds. No personal injury was sustained.

WHAT WENT WRONG:

- Inadequate site control.
- Inadequate maintenance procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Inclement weather condition to be announced beforehand.
- Periodic site and plant inspection to be enforced.
- Unsafe condition reporting system to be enforced.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 25 May 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Concrete fragments broke off and fell from the upper part of the compressor hall building. As a result, the concrete fragments crushed the oil fan lines. No one was injured as a result of the incident.

WHAT WENT WRONG:

- Lack of supervision.
- Inadequate site control.
- Inadequate maintenance procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Periodic site and plant inspection to be enforced.
- Unsafe condition reporting system to be enforced.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

RUSSIA & CENTRAL ASIA ONSHORE

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 25 Sep 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

The protective metal cover of the diffuser fan, on the liquid gas air cooler broke off, damaging the air cooler. The equipment was disconnected and stopped by the plant's operating personnel.

WHAT WENT WRONG:

- Inadequate maintenance program.
- Inadequate plant inspection methods.
- Inadequate control of work and risk assessment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Pre-start inspection to be conducted by competent persons.
- Robust maintenance program to be enforced to identify deformed parts before incident happens.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 27 May 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Hot work

NARRATIVE:

Scaffold board decking erected by the contractor to carry out insulation work between the plant's reactors caught fire which was extinguished by staff with a fire extinguisher.

WHAT WENT WRONG:

- Inadequate supervision.
- Lack of contractor management procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Contractor activity monitoring must be enforced.
- Site control and reporting of incidents culture must be improved.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 03 May 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Work authorization
SECONARY LIFE-SAVING RULE: Hot work

NARRATIVE:

During the repair work on the bypass valve of the service area, a gas odour was noticed. The work was stopped, workers were removed from the area, and gas test showed the presence of combustible gas.

WHAT WENT WRONG:

- Inadequate control of work procedure.
- Inadequate risk assessment.
- Lack of supervision.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Before starting any activity in hazardous area permit to work must be obtained and risk control measures implemented.

RUSSIA & CENTRAL ASIA ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 19 May 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Hot work
SECONARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

While contractor was performing welding work in the vacuum column of the plant, the coke remaining on the column wall smoked from the welding heat.

WHAT WENT WRONG:

- Inadequate control of work procedure.
- Inadequate risk assessment.
- Lack of contractor management procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Area must be inspected, and all combustible materials must be removed before starting any hot work activity.
- Contractor management procedure must be enforced.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 26 May 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Unspecified – other
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

A fire was observed near an oil pipeline in public area. The Fire Department was called to the scene and the fire was extinguished.

WHAT WENT WRONG:

- Lack of supervision,
- Lack of remotely operated monitoring devices.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Methods involving use of monitoring devices, cameras, detectors must be implemented.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 28 Aug 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Exposure electrical
ACTIVITY: Excavation, trenching, ground disturbance
PRIMARY LIFE-SAVING RULE: Work authorization
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While the excavator operator was levelling the ground around the well, the excavator blade damaged and broke the insulation layer of the underground powered electrical cable and as a result of a short circuit, the cable protection in the distribution board was activated.

WHAT WENT WRONG:

- Inadequate control of work procedure.
- Inadequate risk assessment.
- Lack of supervision.
- Lack of training and communication.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Before starting any ground disturbance area layout and location of underground services must be obtained and work mast be authorized by site authority.

CAUSAL FACTORS:

RUSSIA & CENTRAL ASIA ONSHORE

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 09 Feb 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A few minutes after the hydrogen-containing gas compressor of the diesel hydrotreating unit was started, the valve retainer of one of the compressor's load control valves came loose. As a result, the dislodged retainer hit the opposite side and fell to the ground. The compressor was stopped in an emergency and appropriate measures were taken.

No personnel were injured, and no damage was done to the surrounding equipment.

WHAT WENT WRONG:

INADEQUATE MAINTENANCE PROGRAM.

- Inadequate control of work procedure.
- Inadequate risk assessment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Periodic maintenance program to be enforced.
- Proper supervision and authorization of works to be enforced.
- Risk assessment and control of works procedure to be enforced.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 23 Dec 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Struck by (not dropped object)
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Third party vehicle lost control, broke a section of the side of a concrete barrier placed on the side of the public road, and came to rest on a reserve oil line. The oil pipeline was not damaged as a result of the incident.

WHAT WENT WRONG:

Ineffective protective barriers.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Pipelines that cross public area should be protected by more effective barriers.

CAUSAL FACTORS:
PEOPLE (ACTS): Following Procedures: Work or motion at improper speed
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

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DATE: 02 Sep 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Unspecified – Other
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Hot work

NARRATIVE:

During windy weather a fire broke out in the territory of the oil and gas production area. The Fire Department was informed about the fire. The fire was localized and completely extinguished by the fire brigade that arrived at the scene. No damage was sustained. The cause of the fire was not identified.

WHAT WENT WRONG:

- Lack of supervision.
- Inadequate site control.
- Inadequate control of work procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Control of work procedure to be enforced.
- During site inspections potential open flame sources must be identified and controlled.

CAUSAL FACTORS:
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication

RUSSIA & CENTRAL ASIA ONSHORE

PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 18 Nov 2024
COUNTRY: Azerbaijan
FUNCTION: Construction
CAUSE: Explosion, fire or burns
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Hot work
SECONARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

The workers of the contractor company, who were carrying out the work of dismantling the valves in the compression area, were initially loosening the bolts with an open flame, despite the fact that cold work was permitted in the area.

WHAT WENT WRONG:

- Inadequate contractor management procedure.
- Lack of supervision.
- Lack of competency.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Contractors working on site to be more closely monitored.
- Awareness sessions to be arranged for involved workers.

CAUSAL FACTORS:
PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 27 May 2024
COUNTRY: Azerbaijan
FUNCTION: Construction
CAUSE: Falls from height
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Working at height

NARRATIVE:

A decking of scaffold platform erected by contractors to demolish concrete slabs failed and collapsed. The contractor working on the 12-meter-high scaffold platform fell down. Fall arrest system was activated, and the worker remained suspended waiting for rescue. No injury was sustained.

RUSSIA & CENTRAL ASIA ONSHORE

WHAT WENT WRONG:

- Inadequate contractor management procedure.
- Inadequate work planning.
- Inadequate scaffold inspection methods.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Contractor management procedure to be enforced.
- All scaffolds erected must be inspected periodically by competent persons.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 04 Dec 2024
COUNTRY: Azerbaijan
FUNCTION: Construction
CAUSE: Falls from height
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

In order to unload a 12x3 metre container, the worker climbed onto the roof to attach the hook to the container. When he was attaching the hook of the four-legged sling, the place where he was standing (part of the roof) suddenly broke off. The worker fell into the opening and, supporting himself with two elbows, remained hanging, protecting himself from falling and was rescued afterwards. No injury was sustained.

WHAT WENT WRONG:

- Inadequate work planning.
- Lack of supervision.
- Inadequate risk assessment and control of works.
- Lack of competency.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Proper work planning.
- Risk assessment and control of works procedure to be enforced.
- Competency training program to be enforced.

RUSSIA & CENTRAL ASIA ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PEOPLE (ACTS): Use of Protective Methods: Personal Protective Equipment not used or used improperly
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 08 Sep 2024
COUNTRY: Azerbaijan
FUNCTION: Unspecified
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Working at height
SECONARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

The contractor industrial climbers involved in the cleaning of the facade of the administrative building did not properly prepare the work for the cleaning of the facade of the building. The sharp edges of the two-beam beam selected as the anchor point for the industrial climbers' safety rope were not covered with soft material to protect the rope.

Due to the ineffectiveness of measures against the risk of falling objects during work at height, a container containing cleaning fluid fell from the work basket to the ground, into the zone barriered with safety tape.

WHAT WENT WRONG:

- Inadequate contractor management procedure.
- Inadequate control of work and risk assessment.
- Lack of supervision.
- Improper work planning.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Contractor management procedure to be enforced.
- Pre-task workplace inspection to be conducted by work supervisor to ensure anchor points are suitable.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

RUSSIA & CENTRAL ASIA ONSHORE

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 14 May 2024
COUNTRY: Azerbaijan
FUNCTION: Unspecified
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire
SECONARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

A mechanic lifted the cab of a truck and tried to start maintenance work without fully installing the guard under the cab, and the cab fell down. The mechanic just managed to run away.

WHAT WENT WRONG:

- Poor awareness mechanical energy hazards.
- Inadequate control of work procedure.
- Inadequate risk assessment.
- Lack of training.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Awareness sessions to be conducted about potential and gravity energy hazards.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 18 May 2024
COUNTRY: Azerbaijan
FUNCTION: Unspecified
CAUSE: Dropped objects
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

Mechanic used a hydraulic jack to support the vehicle and tried to start repair work beneath the car. Jack failed and mechanic just managed to get aside.

WHAT WENT WRONG:

- Inadequate supervision.
- Inadequate work planning.
- Inadequate maintenance procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Use jack stands: Never rely solely on a hydraulic jack to support the vehicle. Use properly rated jack stands to securely hold the car in place while you work underneath it.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 18 Mar 2024
COUNTRY: Azerbaijan
FUNCTION: Unspecified
CAUSE: Pressure release
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

A mechanic changed the tyres of a truck and while testing one of the tyres burst. No injury was sustained.

RUSSIA & CENTRAL ASIA ONSHORE

WHAT WENT WRONG:

- Inadequate maintenance practices.
- Lack of supervision.
- Inadequate vendor control procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- All spare parts should be provided by vendors that are screened and monitored by the company.
- All maintenance works must be authorized by competent persons and approved spare parts must be used only.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 13 Aug 2024
COUNTRY: Azerbaijan
FUNCTION: Unspecified
CAUSE: Struck by (not dropped object)
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While loading cargo onto a truck using a mobile crane, the rear left outrigger failed due to uneven ground causing the cargo to lose balance and touching nearby fence.

WHAT WENT WRONG:

- Inadequate work planning.
- Inadequate supervision.
- Lack of competency.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Competency training program to be enforced.
- Control of Work procedure to be enforced.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper lifting or loading

PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

RUSSIA & CENTRAL ASIA OFFSHORE

DATE: 08 Oct 2024
COUNTRY: Azerbaijan
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While the drilling tower was being pushed 3 metres from north to south by a 1.25 metre long hydraulic jack to a specified point, 1 and 1.5 metre conductors were placed between the jack and the drilling tower. During the pushing, the foreman observed that the conductors were displaced and rose up and he stopped the work. Then, when he wanted to ensure the regular arrangement of the conductors by pulling the jack rod back using the hydraulic remote control, the 1 metre conductor slipped and hit the jack rod, and as a result, the jack fell into the area where the wells were located through the gap formed in the floor during the pulling of the tower. As a result of the incident, no damage was caused to people, equipment, or production wells.

WHAT WENT WRONG:

- Inadequate work planning.
- Inadequate risk assessment.
- Inadequate control of work procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Proper work planning involving any unexpected situations.
- Risk awareness sessions involving all crew members.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 06 Sep 2024
COUNTRY: Azerbaijan
FUNCTION: Drilling
CAUSE: Struck by (not dropped object)
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

RUSSIA & CENTRAL ASIA OFFSHORE

NARRATIVE:

During the replacing of pipes of various sizes by a crane on the offshore platform deck, a 244.5 mm pipe first hit the material warehouse and then the tower stairs due to the lack of a tagline attached to the load. As a result of the incident, no damage was caused to people, equipment, or production wells.

WHAT WENT WRONG:

- Lack of supervision.
- Inadequate work procedure.
- Inadequate risk assessment.
- Lack of competency.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Competency training program to be enforced.
- Control of Work system to be revised.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 31 May 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

The connecting lines between the piles failed in the lower part of the offshore platform with 25 cubic metre fuel tank located on the deck. The platform did not collapse. There were 10 cubic metres of fuel inside the 25 cubic metre fuel tank.

WHAT WENT WRONG:

- Lack of supervision.
- Inadequate site control.
- Inadequate maintenance procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Periodic site and plant inspection to be enforced.
- Unsafe condition reporting system to be enforced.

RUSSIA & CENTRAL ASIA OFFSHORE

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision

DATE: 21 Dec 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Hot work
SECONARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

On the offshore platform an oxygen cylinder was observed with red hose (intended for fuel gas) attached.

WHAT WENT WRONG:

- Lack of supervision.
- Lack of competency.
- Inadequate control of work procedure.
- Lack of risk awareness.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Competency training program to be enforced.
- Control of work procedure to be enforced.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

RUSSIA & CENTRAL ASIA OFFSHORE

DATE: 20 Sep 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Falls from height
ACTIVITY: Transport – Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

While the worker was getting off the support vessel onto the boarding platform of the offshore platform, he tripped and fell towards the sea. At the last moment, a rescue sailor on the boarding platform grabbed the worker's clothes and pulled him onto the platform.

WHAT WENT WRONG:

- No safe transfer methods, gangway, or transfer bridge was provided.
- Inadequate protective barriers.
- Lack of supervision.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Safe transfer methods, gangway, or transfer bridge to be provided.
- Awareness session to be arranged for all crew members.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress
PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 21 Sep 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

The 5 slab plates of the offshore trestle bridge collapsed. In that area there was a 2-inch gas lift line, a processing line, an 8-inch oil and gas line, and a 12-inch technical water line running through. No personnel were on the spot.

WHAT WENT WRONG:

- Lack of supervision.
- Inadequate site control.
- Inadequate maintenance procedure.

RUSSIA & CENTRAL ASIA OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Periodic site and plant inspection to be enforced.
- Unsafe condition reporting system to be enforced.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Inadequate surfaces, floors, walkways or roads

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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DATE: 20 Oct 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

NARRATIVE:

Due to windy weather conditions, a 50m2 section of the offshore trestle way collapsed. 12-inch oil and gas pipelines from the offshore platforms to the oil storage facility passed through the area, and a 5m section of the pipeline remained suspended over the water. The suspension of the pipeline did not cause any complications in the production process. No personnel were on the spot during incident.

WHAT WENT WRONG:

- Lack of supervision.
- Inadequate site control.
- Inadequate maintenance procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Periodic site and plant inspection to be enforced.
- Unsafe condition reporting system to be enforced.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Work Place Hazards: Storms or acts of nature

RUSSIA & CENTRAL ASIA OFFSHORE

DATE: 16 May 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Water related, drowning
ACTIVITY: Transport – Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Work authorization
SECONARY LIFE-SAVING RULE: Bypassing safety controls

NARRATIVE:

Contractor ship with company passengers on the deck ran aground 8.3 km from the coast. Support ships were immediately deployed to the area and the passengers were evacuated ashore.

WHAT WENT WRONG:

- Inadequate planning.
- Lack of contractor management procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- If there is any deviation from the established route, the master should get authorization.
- Contractor audit procedure to be implemented.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication

DATE: 11 Nov 2024
COUNTRY: Azerbaijan
FUNCTION: Production
CAUSE: Water related, drowning
ACTIVITY: Transport – Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Work authorization

NARRATIVE:

The contractor ship with company workers on the deck ran aground near the offshore platform. The support vessel was assigned to remove the ship from the sand. By the time the support vessel arrived, the ship itself had left the sand and was moored at the pier.

WHAT WENT WRONG:

- Inadequate planning.
- Lack of communication.
- Lack of contractor management procedure.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- The ship master must get authorization prior any deviation from the route.
- Contractor audit process to be established.

RUSSIA & CENTRAL ASIA OFFSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate communication

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DATE: 09 Oct 2024
COUNTRY: Azerbaijan
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the installation of a 24m long pile with a crawler crane, the length of the pile did not match the crane arm, so the work was stopped due to the possibility of the pile pipe hitting the crane limiter and other structures on the deck of offshore platform.

WHAT WENT WRONG:

- Inadequate work planning.
- Inadequate work procedure.
- Lack of supervision.
- Lack of competency.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Competency training program to be enforced.
- Control of Work procedure to be enforced.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment
PROCESS (CONDITIONS): Organizational: Inadequate supervision
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 07 Dec 2024
COUNTRY: Argentina
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the final stage of connecting the tubing hanger to the well head, a 3 ton object fell due to the failure of the lifting sling.

WHAT WENT WRONG:

- Work context – within the operating procedure the task was not considered critical (sling breaks due to overload).
- Unsafe acts – personnel was in the line of fire.
- Failed safeguards – during task planning, the risk analysis was not reviewed.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Implementation of the pre-operational checklist for the task.
- ATS update of the "packaged with TTL" manoeuvre.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Following Procedures: Improper lifting or loading
PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 15 Dec 2024
COUNTRY: Argentina
FUNCTION: Drilling
CAUSE: Pressure release
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During a pressurization operation, a section of the connection detached and struck the trailer.

WHAT WENT WRONG:

Organizational Factor:

- Unclear definition regarding who should be performing the test on this section.
- Miscommunication between the two contractor companies involved.

Work Context:

- The traceability of the failed materials was missing.
- The companies did not test the section that broke off.

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Unsafe Acts:

- The pumping operation began without opening the frac stack valve.

Failed Defences:

- Communication and coordination between the manoeuvres and contractors was not considered in any risk analysis.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Incorporate into operating procedures and risk analysis interaction with third parties and companies.
- Develop a procedure for setting and testing shut down switches and relief valves.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 04 Aug 2024
COUNTRY: Argentina
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Excavation, trenching, ground disturbance
PRIMARY LIFE-SAVING RULE: Work authorization
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During an excavation with a backhoe to uncover a service rig anchor at the well pad, a machine operator perceived contacting an underground object at about 0.6 m depth, stopping immediately the operation.

When inspecting the area closer, the operator observed a scratch of the protective coating of an underground flexible production pipe. The operator turned off the machine, and alerted the testing operator and they both moved to a safe distance to call the company oil spill response team.

While waiting for an answer, they observed fluid coming out of the trench initially in the form of spray, which caused a spill of crude oil, the projection of some rocks that caused damage to the windshield of the backhoe. All the wells on the pad were shut in and left in safe condition. There was no harm to people.

WHAT WENT WRONG:

Design issues:

- No warning mesh installed as per design requirement.
- Flex pipe too close to the surface, not meeting design depth.
- No sand fill was left as per engineering requirement.
- Distance between anchor and Flex pipe very close in original layout.

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Procedures/PTW:

- Contractor worked under a generic permit to work (PTW), not addressing excavations.
- Contractor did not have an excavation procedure.
- Company required simultaneous operations (SIMOPS) controls not implemented.
- No access control at the well pad (no lock and/or authorization).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

There were gaps in procedures (PTW/SIMOP) not captured previously by assurance system. A similar incident occurred in the past and no corrective actions were taken. Learning from incidents and assurance effectiveness is critical for reducing incident probability.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

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DATE: 17 May 2024
COUNTRY: Argentina
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Hot work

NARRATIVE:

The ingress of a hydrocarbon slug from the producing wells caused the levels in the treatment section vessels to rise and flood. As a result, unstabilised condensate entered the glycol regeneration train. Finally, the equipment in the glycol regeneration section was flooded and oil was vented through the vent stack. This activated the flammable gas detectors in the surrounding areas and caused the unit to shut down.

WHAT WENT WRONG:

The need to simultaneously bypass multiple layers of protection due to configuration/calibration problems after the initial planned maintenance of the unit, combined with poor risk management, allowed the undetected overfilling of various vessels until condensate venting occurred through the stack of the glycol regeneration section.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

In the case of safety critical equipment inhibition, it is essential to take a holistic view in order to identify multiple inhibits and assess the overall risk level for decision making.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 02 Apr 2024
COUNTRY: Brazil
FUNCTION: Drilling
CAUSE: Dropped objects
ACTIVITY: Transport – Water, incl. marine activity
PRIMARY LIFE-SAVING RULE: Bypassing safety controls
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the weekly preventive maintenance of the alternative lifeboat #4 on the port side, the aft ring broke, leaving the ring open for a few minutes. Shortly after, there was a complete opening of the ring, causing the aft part of the lifeboat to detach, and the forward ring could not withstand the load and broke, leading to the lifeboat falling into the water. No one was injured.

WHAT WENT WRONG:

- Ineffective training: knowledge-based mistake in taking measures to substitute the material. No management of change was developed for the material alteration. Failure in the application of the management of change procedure by the marine team. Low frequency of process usage leading to poor training absorption.
- Fragile supply management system: failure in the item registration process. Possibility of purchasing an item without an associated asset. Possibility of purchasing a spare part as a consumable. Fragility in the material approval flow. Lack of controls in the computerized systems for the acquisition and disposal of materials.
- Inadequate product management: failure to meet the specifications of the requested item. Certificate issued without conducting load tests.
- Incomplete procedure: failure in planning the replacement of an item without involving the manufacturer. Procedure on lifeboat maintenance does not clearly outline the applicable guidelines for component replacement.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- The management of change (MoC) process for "critical assets" should have a standard action for consulting a competent third party (manufacturer and/or classification society).
- Conduct a workshop for employees on the revised management of change procedure, focusing on material and equipment changes (insights about materials). Implement controls in the system during the procurement request and material disposal process.
- Hire specialized consulting to identify best market practices that can be implemented in supplier management.
- Review the maintenance procedure for lifeboats and boats, clearly outlining the guidelines for item replacement.
- Standardize and review the classification of inventory items, implementing levels of criticality.
- Contract external specialists to conduct internal audits.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PEOPLE (ACTS): Use of Protective Methods: Equipment or materials not secured
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change
PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing
PROCESS (CONDITIONS): Organizational: Inadequate training/competence
PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 08 Jan 2024
COUNTRY: Brazil
FUNCTION: Drilling
CAUSE: Explosion, fire or burns
ACTIVITY: Drilling, workover, well operations
PRIMARY LIFE-SAVING RULE: Hot work
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the well damping operation by bullheading, the pumping pressure reached the configured limit of the relief valve (hydraulic pop-off) of the mud pump, causing its activation and consequently the failure of the primary well barrier envelope. The opening of this valve was not noticed by the team on board and resulted in the loss of hydraulic pressure on the formation. Furthermore, as monitoring parameters were not fully established, as well as a hidden failure in monitoring the pop-off discharge tank, there was a short delay in closing the valves of the subsea workover assembly and Surface Flow Tree (secondary Well barrier envelope), allowing the migration of gas through the production column and DPR (Drill Pipe Riser), reaching the tank room (100% LEL) through the relief valve and, subsequently, discharge on the main deck of the unit through the air exhaust system of the storage room tanks, covering a significant extension of unclassified area (peak of 64% LEL on the sensor close to the well testing plant).

WHAT WENT WRONG:

- Inadequate operating procedure: There was a failure to monitor parameters related to the fluid during the well damping operation.
- Failure in the management and control of critical operational safety elements: There was a late execution of the well closure.
- Failure in the risk identification and analysis methodology: Exhaust from the tank room directed to an unclassified area close to the well test boiler.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Include in the Driller's checklist the verification that the tank sensors are activated.
- Communicate to the teams that during the pumping of fluid, the level of the tank used for pumping must be monitored, that is, tank that must lose the pumped volume, as well as the tank that receives the fluid in case of opening the pop-off and consequently will gain volume in case of opening it.
- Include audible and visual alarm in the driller cabin in case of pop-off valve opening.
- Include questions in the design and operation checklists related to pumping pressure limits in BH (bullheading) operations in LWO (light workover).
- Establish critical operations for on-site monitoring by inspectors, toolpusher and others involved in hydrocarbon risk situations. (Ex: Home well damping, dry test execution, and others.).
- Establish a list of loss of containment evidence for LWO that result in the need for immediate closure of the well. Ex: 1-general alarm of CH4 of unknown origin; 2-Minimum pressure limit on the head at the beginning/during BH; 3-Unexpected return of hydrocarbons at the Well Testing plant and others.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)
PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)
PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

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PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures
PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 02 Feb 2024
COUNTRY: Brazil
FUNCTION: Production
CAUSE: Caught in, under or between (excl. dropped objects)
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

The employee reported to the infirmary along with his supervisor, informing that during the maintenance of the starting air compressor, he suffered an impact from the engine's propeller on the fourth digit of his left hand, sustaining a cutting injury to the first phalanx.

WHAT WENT WRONG:

- Did not consider the reduction of human exposure to the consequences of potential equipment or system failures. The manufacturer did not consider the risk of failure in the fan protection due to construction fragility.
- Inadequate identification, evaluation, consideration, and mitigation of risks. There was no risk management regarding the fragility of the fan protection for maintenance and operation activities.
- Inadequate consideration of layout, human factors, and external causes. The initial design of the unit did not consider the appropriate layout for the activities that would be carried out on that equipment.
- Inadequate provision of resources. Although within the contractual deadline, the contractor did not provide the most appropriate PPE, beyond the minimum required.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Lessons Learned:

- Seek to "procedure" an alternative means of disembarking individuals with medical emergencies in cases of helipad interdiction and crane inoperability. E.g., transfer by basket using UMS crane. Suggestion: Review the emergency response plan to include the alternative so that there is no loss of time with negotiations in similar scenarios.
- Reinforce the need to preserve the accident scene and its evidence.
- Monitor and control adherence to contractual requirements made through external correspondence.

Recommendations:

- It is recommended to check if there are similar protections in Company facilities. If so, correct immediately with a temporary solution.
- Arrange with the compressor manufacturer for a design of fan protection using appropriate mechanical resistance material.
- Provide and install a permanent solution for the fan protections similar to those found in Company facilities.
- Conduct a comprehensive inspection of the protections for rotating equipment in the units regarding low mechanical resistance. Correct if necessary.
- Redirect the oil collection line of starting air compressor No. 2.
- The contracted company should adjust the supply of gloves.

SOUTH & CENTRAL AMERICA OFFSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective Personal Protective Equipment

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 19 Apr 2024
COUNTRY: Brazil
FUNCTION: Production
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the manoeuvre to move a pump from the float on the main deck at the aft port side, the auxiliary cable of the port crane rubbed against a fibre cable tray located above the movement area, causing its cover to detach and fall, hitting the professional involved in the manoeuvre in the lower region, resulting in a blunt cut on his left eyebrow.

WHAT WENT WRONG:

- Contract Transition.
- Situation or condition not anticipated in the planning.
- Workplace layout.
- Inadequate identification and qualitative or quantitative analysis of risks.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Lessons Learnt:

- Importance of mapping all elements with potential for falling in the risk inspection plan.
- Conduct workplace assessments to ensure the absence of obstacles before load operations.
- Monitoring of new employees in the unit and ensuring the presence of experienced supervisors during contract transitions.
- Conduct pre-work meetings for all activities, ensuring discussion and understanding of all potential risks.

Recommendations:

- Establish with the contract management the onboarding of the supervisor (loading team) from any new company providing services to the unit, to oversee the routine and specifics of platform operations.
- Provide training to the loading supervisor about the responsibilities of their role, which should plan operations by assessing the chosen location, ensuring it is free of obstacles.
- Adapt the preliminary risk analysis document to a format consistent with standards for Load Handling.
- Provide training to the loading supervisor to conduct a pre-work meeting for the load handling team, even for routine movements.
- Suggest to the manager of standard that the need for an assessment based on standard be extended to routine activities.

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- Evaluate/implement alternative methods for load handling in the aft port main deck area to ensure safe manoeuvring.
- Assess whether other areas of the platform also have space limitations that could compromise load handling manoeuvres.
- Include all cable trays in the object fall prevention inspection plan.
- Install reinforcement fixtures on all cable trays near load handling areas.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

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DATE: 23 Oct 2024
COUNTRY: Brazil
FUNCTION: Construction
CAUSE: Dropped objects
ACTIVITY: Lifting, crane, rigging, deck operations
PRIMARY LIFE-SAVING RULE: Safe mechanical lifting
SECONARY LIFE-SAVING RULE: Line of fire

NARRATIVE:

During the operation of installing the offloading line using the forward reel of a vessel, the bearing of the chain that moves the forward reel's spooling laterally broke, causing the bearing and chain assembly (weighing more than 100 kg) to fall to the deck floor from an approximate height of 3 metres, posing a risk of injury or fatality.

WHAT WENT WRONG:

- Failure in the hiring of suppliers and acquisition of components.
- Failure in the system design.
- Failure in project monitoring.
- Failure in the operation procedure.
- Failure in maintenance and inspection.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

It is possible to identify the need for improvement in the supplier hiring process and acquisition of components, especially those of a critical nature. An area of concern is related to the company's documentation management, which was insufficient in several aspects. The committee also identified opportunities for improving the spooling project to be evaluated by the shipowner.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 01 Mar 2024
COUNTRY: Trinidad and Tobago
FUNCTION: Production
CAUSE: Pressure release
ACTIVITY: Maintenance, inspection, testing
PRIMARY LIFE-SAVING RULE: Work authorization
SECONARY LIFE-SAVING RULE: Energy isolation

NARRATIVE:

During a routine field visit, it was identified that a 2-inch pipe spool associated with the Low-Pressure Flare Header was installed, which sits outside the boundary isolation, as per approved isolation plan. The spool piece in question appears to have been installed during the removal and replacement of piping scope of work on the glycol regeneration system. However, this was not part of the approved breaking of containment scope of work.

WHAT WENT WRONG:

- Work planning was ineffective (multiple layers).
- Scope changes not managed effectively (multiple changes).
- Change of Work Planning not managed.
- Change of Work Execution/Monitoring not managed effectively.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Safe work execution requires adherence to process, work scope change management, effective communication and operational discipline.
- Safe to Speak up is fragile, less present than everyone believes and significantly informs the organizational culture.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)
PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)
PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement
PROCESS (CONDITIONS): Organizational: Inadequate communication
PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

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For the full analysis of 2024 results, as well as all fatal incident reports submitted to IOGP since 1991, visit IOGP's data website at: https://data.iogp.org

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